

Seismic performance of self-centering prestressed concrete frame with auxetic perforated steel plate shear walls

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ABSTRACT

This study examines the seismic behavior of an innovative self-centering prestressed concrete (SCPC) frame system that integrates auxetic steel plate shear walls featuring peanut-shaped perforations (AP-SPSWs). The system's prefabricated components are designed for easy replacement, enhancing both the efficiency and practicality of the structure. The inclusion of AP-SPSWs enhances the structure's energy dissipation, enables streamlined integration of pipelines, and reduces the overall structural weight. To assess the seismic performance, three specimens with varying geometries of steel plates were designed and manufactured. These specimens underwent cyclic loading tests, and a finite element (FE) model was subsequently developed and validated based on the experimental data. The results demonstrated that the precast beam and column components and post-tensioned (PT) tendons remained largely elastic, while the thin steel web plates experienced plastic deformation under loading. Significant plastic deformation in the auxetic perforated steel infill panels contributed to the system's energy dissipation. The PT tendons facilitated a robust self-centering capability in the specimens. Following this, a thoroughly parametric analysis was conducted and revealed that greater thickness and a higher r_1/D ratio resulted in enhanced energy dissipation and stiffness. Additionally, increasing the initial PT force contributed to enhanced stiffness and strength of the AP-SCPC systems.

1. Introduction

Earthquakes, as a common natural disaster worldwide, have caused significant damage and extensive residual deformation to building structures, often making them challenging or even impossible to recover [1–4]. To mitigate the substantial economic losses caused by earthquakes, it is essential to enhance the seismic resistance and post-earthquake reparability of buildings. In response to this need, self-centering prestressed concrete (SCPC) frame structures have been developed as an innovative seismic-resistant system aimed at improving structural resilience by minimizing residual drift during seismic events [5].

As shown in Fig. 1, the SCPC frame structure differs from traditional reinforced concrete frames in that its beams and columns are connected by post-tensioned (PT) strands. Previous research has demonstrated that SCPC frames can achieve adequate strength and ductility while undergoing large inelastic displacements, all while maintaining excellent self-centering capacity [6–8]. However, a primary concern for SCPC frames their potential for excessive lateral displacements when subjected to strong earthquake ground motions due to limited energy dissipation [9]. To address this issue, various energy-dissipating devices have been integrated into SCPC

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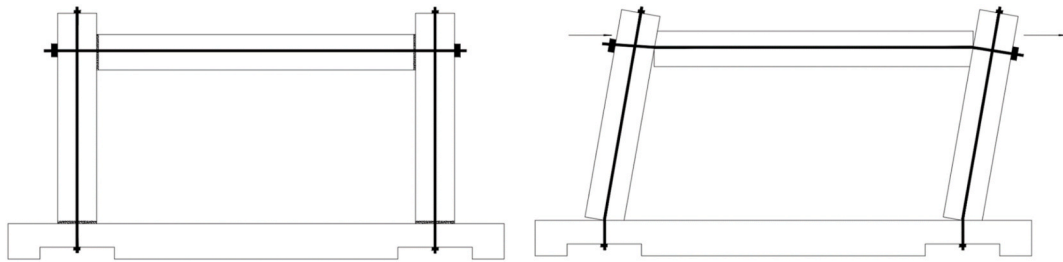


Fig. 1. Schematic illustration of SCPC frame structure and its deformation pattern.

frames to enhance their energy absorption and dissipation capacities [10–17]. During earthquakes, these devices localize damage, while the prestressing tendons' self-centering capability helps limit permanent deformations. Consequently, structures can be efficiently repaired and restored, enhancing seismic resilience and supporting the development toward construction industrialization.

In recent years, numerous studies have focused on the behavior and design of seismic-resistant self-centering precast concrete (SCPC) frames. Song et al. [9,18,19] proposed a precast post-tensioning beam-column connection, where the web friction devices were adopted in the connection region as the energy dissipators. Their experimental results indicated that the beam was the only member that experienced slight damage, while the column remained undamaged up to 2.5 % lateral drift. Lu et al. [20] introduced a three-dimensional precast reinforced concrete (RC) frame assembled using post-tensioned tendons in three directions, testing it under quasi-static loads, and demonstrated a strong deformation capacity with limited residual deformations. Yang et al. [21] presented a novel self-centering precast steel-concrete composite frame, analyzing its seismic performance. Their study showed that the frame effectively managed residual deformations during the self-centering phase and dissipated energy during the energy absorption phase. Cai et al. [22,23] performed systematic investigations into the behavior of a new prestressed beam-column connection in precast concrete structures subjected to quasi-static loading and reparability upon unloading. The results suggested that as the energy dissipators and connecting devices, the steel angles offered the majority of energy dissipation in the subassemblies through extensive repeated plastic deformation.

Infill plate walls can enhance the initial stiffness and energy dissipation capacity of SCPC frames while increasing their residual deformation [24–26]. Due to their exceptional ductility and energy dissipation capabilities [27–29], steel plate shear walls (SPSWs) have been integrated as energy-dissipating components not only in steel building structures, but also in concrete structures [30–37]. The experimental research on steel plate shear walls (SPSWs) connected through unbonded post-tensioning and observed that these systems exhibit high levels of ductility, initial stiffness, and significantly higher energy dissipation than anticipated [30–33]. The seismic response of SC-SPSWs equipped with slit steel plate shear walls was investigated [34–36]. Their findings showed that the steel slit panel frame exhibits good energy dissipation and high ductile behavior, its lateral stiffness and ultimate bearing capacity can be adjusted relatively and independently by changing the slits parameters. Dowden and Bruneau [37] analyzed the cyclic behavior of self-centering SPSWs with various distinct boundary element connections and configurations of infill panels. Their results revealed that self-centering SPSWs with solid web plates provide superior stiffness and energy dissipating capacity over those with web strips, though the self-centering ability of those with solid infill plates was slightly diminished.

Cutting materials out from the solid web have been a popular and effective approach to enhance architectural functions of SPSWs since a few decades [38,39], though reduced load capacity and stiffness are expected. Consequently, the geometries and arrangements of these cutouts have gained considerable interest due to their critical influence on the overall properties of SPSWs. Inspired by the unique characteristics of auxetic materials, known for their negative Poisson's ratio and exceptional shear resistance and energy absorption capabilities, researchers have proposed auxetic steel plate walls by incorporating orthogonal perforations [40]. Wang et al. [40] introduced an innovative steel plate shear wall with auxetic-shaped cutouts (AP-SPSW) featuring peanut and elliptical shapes.

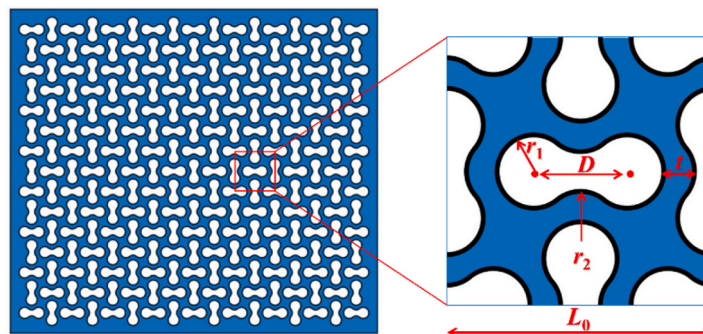
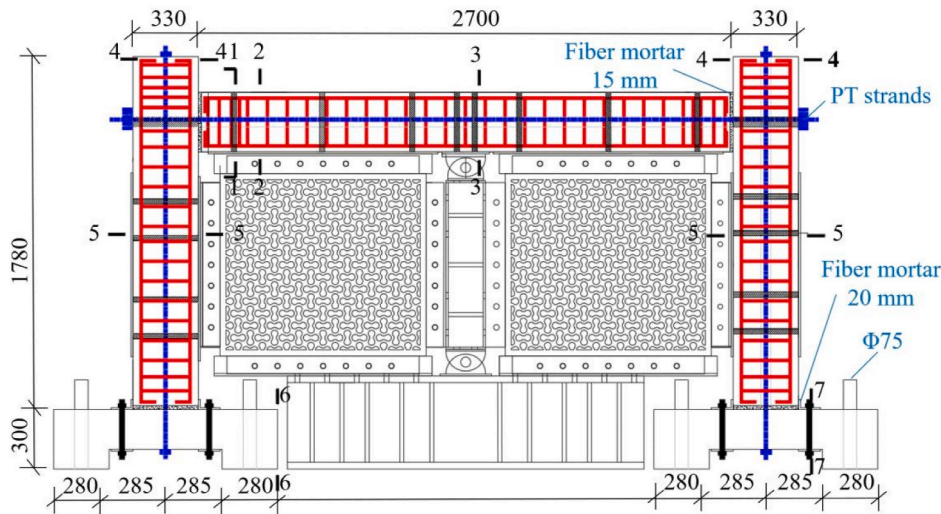


Fig. 2. Details of AP-SPSW

Table 1

Geometric parameters of auxetic perforated steel plate shear wall.

Specimens	r_1/mm	r_2/mm	D/mm	t/mm	t_w/mm	Φ	r_1/r_2	r_1/D
OP-1-0.4	17	17	43	14.03	3	47.6 %	1.0	0.4
OP-0.85-0.4		20	43	6.75		64.1 %	0.85	0.4
OP-1-0.55		17	30	16.15		42.9 %	1.0	0.55

**Fig. 3.** View of AP-SCPC specimen.

Their study demonstrated that this design effectively reduces out-of-plane instability deformation, with the auxetic deformation mechanism contributing significantly to improved shear buckling resistance. Although with good seismic performance in previous practices, the perforated steel plate shear walls have rarely been used in concrete building structures. Motivated by this, this study attempts to combine the idea of self-centering precast concrete frame and the newly developed AP-SPSWs. More specifically, the AP-SPSWs with peanut-shaped cutouts have been implemented in SCPC frames, serving as efficient energy-dissipating devices to enhance seismic resilience.

This study investigates the seismic performance of precast prestressed concrete frame structures incorporating AP-SPSWs (AP-SCPC) through both experimental and numerical approaches. Three groups of cyclic loading tests were performed to evaluate the system's seismic behavior. Following the experiments, detailed finite element models were developed and verified based on the test results. The validated models were subsequently utilized to examine the effects of critical design parameters such as the thickness of the web plate (t_w), the ratio of the head to side circle radii (r_1/r_2), the aspect ratio of the peanut-shaped cutout (r_1/D) in AP-SPSW, and the initial post-tensioning (PT) prestress (f_{pi}/f_{ptk}) in precast beam and columns on the seismic performance of the AP-SCPC system.

2. Experimental program

2.1. Detailing of the specimens

Three frame subassembly specimens with distinct AP-SPSWs shown in Fig. 2 and outlined in Table 1 were subjected to cyclic loading tests. The AP-SPSWs were different in the geometry of the cutouts. The layout of the AP-SCPC specimen is presented in Fig. 3. A two-dimensional frame, featuring a single-story and single-span configuration, was developed as a 0.5-scale subassembly. The test specimens were designed based on the first story of a 5-story prototype self-centering concrete frame structure by Cai [41]. The experimental setup included a self-centering prestressed concrete frame structure subassembly with AP-SPSWs connecting through a vertical element. Considering that the introduction of perforations in AP-SPSWs may reduce their contribution to the overall load-carrying capacity of AP-SCPC compared to conventional SPSWs, two AP-SPSWs were employed to ensure sufficient strength for the AP-SCPC. The bare frame subassembly was reused during the experimental program.

The self-centering prestressed concrete (SCPC) system consisted of a precast beam, two precast columns, and a reinforced concrete (RC) foundation. Fig. 4 provides specific measurements for these components. The beam spans 2670 mm, with a cross-sectional width of 180 mm and a depth of 300 mm. Each column is 1760 mm tall, with cross-sectional dimensions of 280×330 mm, while the foundation has a depth of 300 mm and measures 500×1150 mm in cross-section. A groove, 100 mm deep and measuring 300×570 mm in cross-section, was incorporated to allow for the connection between the post-tensioning (PT) strands and the anchors. Additionally, tendon ducts with diameters of 50 mm for PT strands, 25 mm for high-strength bolts, and 75 mm for anchors were installed, as

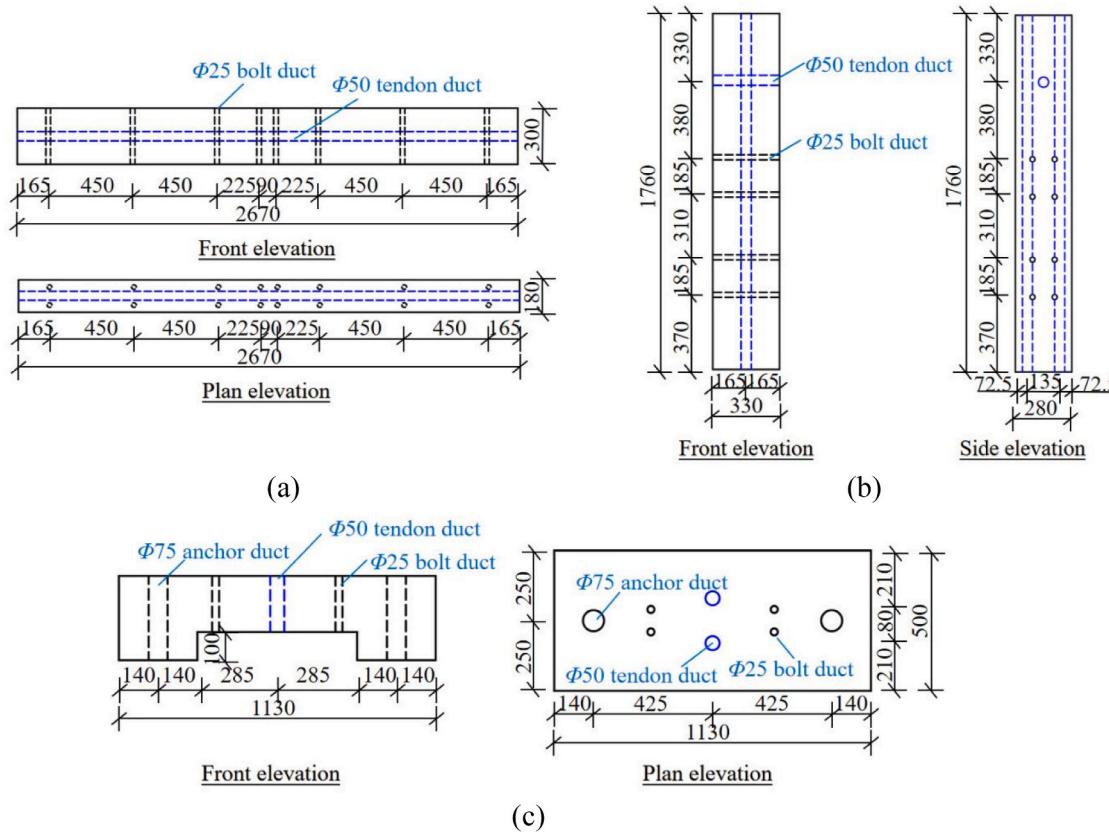


Fig. 4. Details of precast members, (a) beam, (b) column, and (c) RC foundation.

illustrated in Fig. 4. The reinforcement layout for all precast RC elements is depicted in Fig. 5, with a cover of 10 mm applied to the beam, columns, and foundation. The AP-SPSWs were connected to the precast beams and columns through boundary elements. Each infill plate of the AP-SPSWs had in-plane dimensions of 1040×920 mm, with a central region measuring 980×857.5 mm, which was perforated to form the auxetic structural configurations. The four edges of the infill plate, each 60 mm wide, were reserved for the boundary element connections. Further details on the perforations are provided below.

The arrangement of unit cells within the AP-SPSW, as illustrated in Fig. 2, is a key factor in generating the auxetic effect in the structure with negative Poisson's ratio. The unit cells with multiple orthogonal perforations of peanut shape were designed following the method introduced by Wang et al. [40], Ren et al. [42], and Xin [43]. The indicator of perforation shape controls both Poisson's ratio of the unit cells and the mechanical behavior. As recommended by Wang et al. [40], the indicator corresponding to the porosity between 40 % and 70 % was chosen to ensure the unit cells with negative Poisson's ratios and that the entire AP-SPSW achieves desired strength and ductility. The geometric characteristics of the AP-SPSWs are presented in Table 1, where parameters r_1 , r_2 , D , and Φ represent the radii of the head circles, the side circles, the distance between the head circle centers, and the porosity of each unit cell, respectively. The notation "OP" in the design refers to the AP-SPSW, with the following numbers specifying the ratios of r_1/r_2 and r_1/D . Three specimens were designed, each with an identical web plate thickness of 3 mm. The key design ratios were $r_1/r_2 = 1.0$ and 0.85 , and $r_1/D = 0.4$ and 0.55 . The porosities (Φ) of the perforated cells were 47.6 %, 64.1 %, and 42.9 %, respectively. The porosity (Φ) is calculated as follows:

$$\Phi = \frac{4A_p}{L_0^2} \quad (1)$$

where $L_0 = \sqrt{4(r_1 + r_2)^2 - D^2} + 2(r_2 - r_1 + t) + D$ is the side length of the perforated square unit cell, t is the minimum thickness of the ligament between two neighboring peanuts, and A_p is the area of the peanut shape calculated as:

$$A_p = 2\pi r_2^2 + \frac{DS}{2} - 2r_1^2 \arcsin\left[\frac{S}{2(r_1 + r_2)}\right] - 2r_2^2 \arccos\left[\frac{S}{2(r_1 + r_2)}\right] \quad (2)$$

where $S = \sqrt{4(r_1 + r_2)^2 - D^2}$.

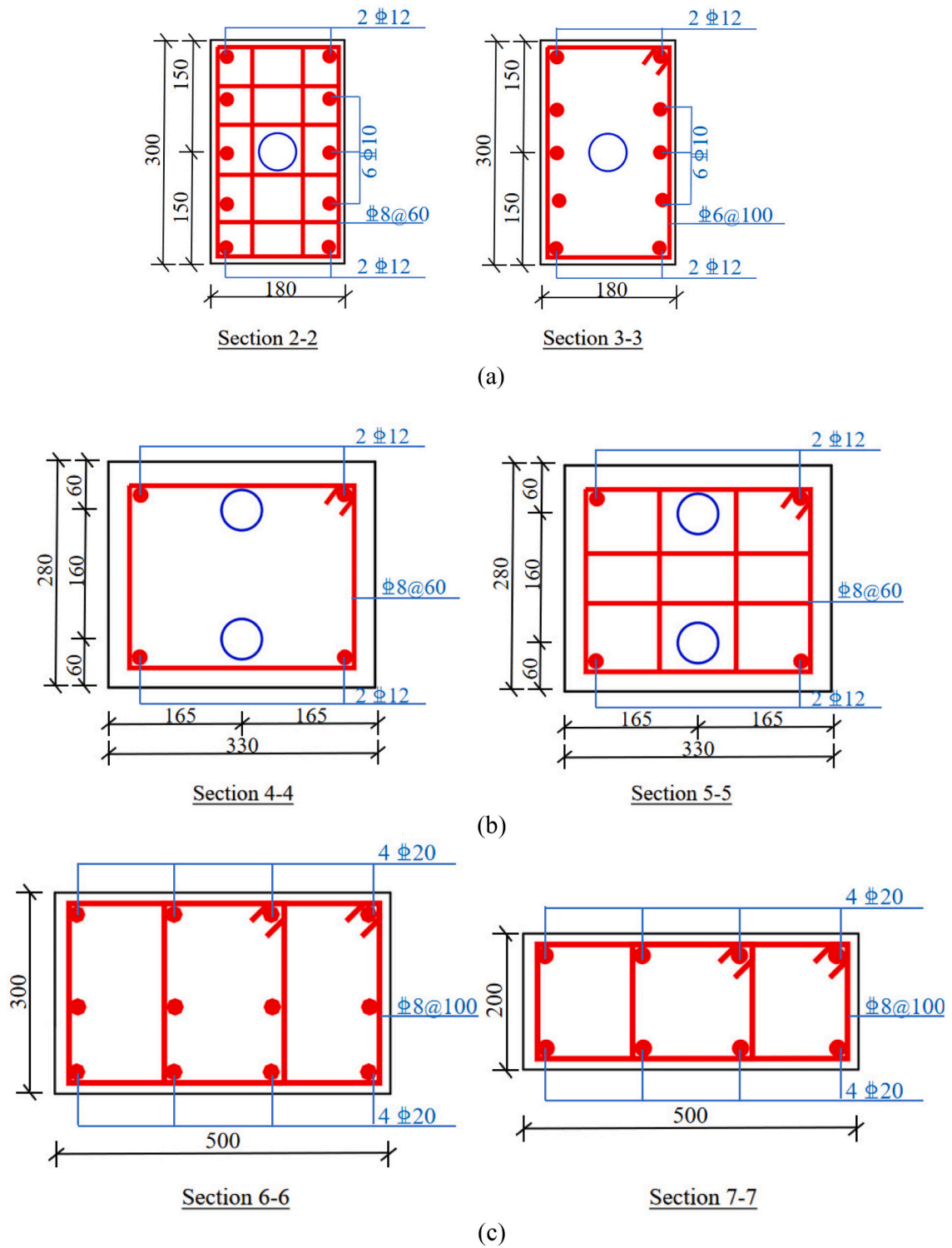


Fig. 5. Detailing of precast members, (a) beam, (b) columns, and (c) RC foundations.

The connection device, as shown in Fig. 6, consists of a horizontal boundary element, a vertical boundary element, a steel I-column, a hinged element, and a steel foundation. The horizontal and vertical boundary elements (HBEs and VBEs) are fabricated using Chinese equal-angle steel sections (L100 mm $\times$ 100 mm $\times$ 12 mm). The two legs of the angle sections are provided with holes of different sizes.

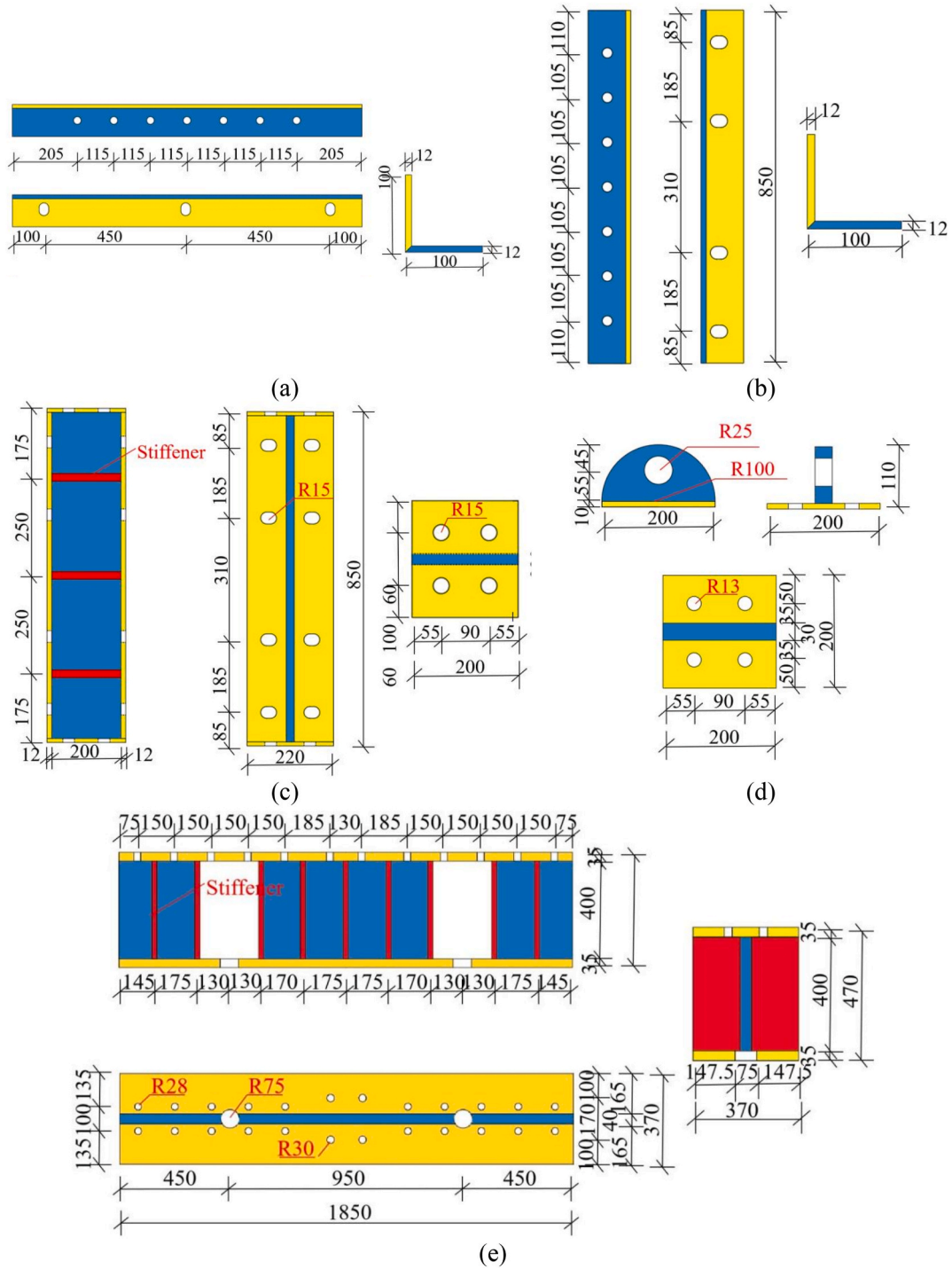


Fig. 6. Details of connection device, (a) HBE, (b) VBE, (c) H-shaped steel column, (d) hinged element, and (e) steel foundation.

One leg is machined with seven 24 mm diameter bolt holes to connect to the AP-SPSW, while three, four, and five elliptical bolt holes were machined for connections with the precast beam, steel I-column, and steel foundation, respectively. The elliptical shapes for the bolt holes were set for the purpose of mitigating installation errors.

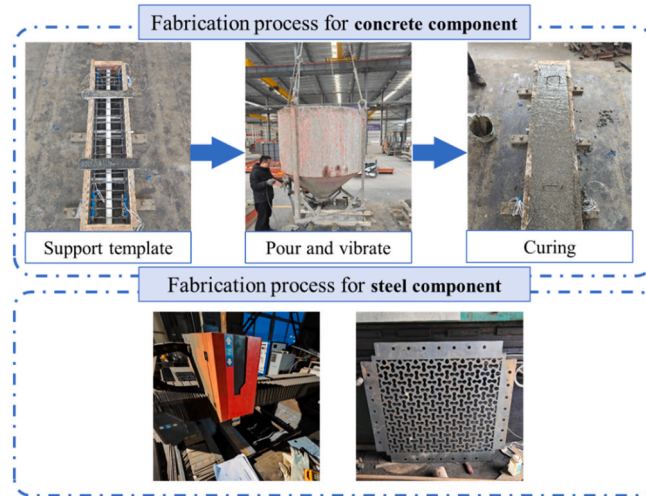


Fig. 7. Fabrication process for concrete and steel components.

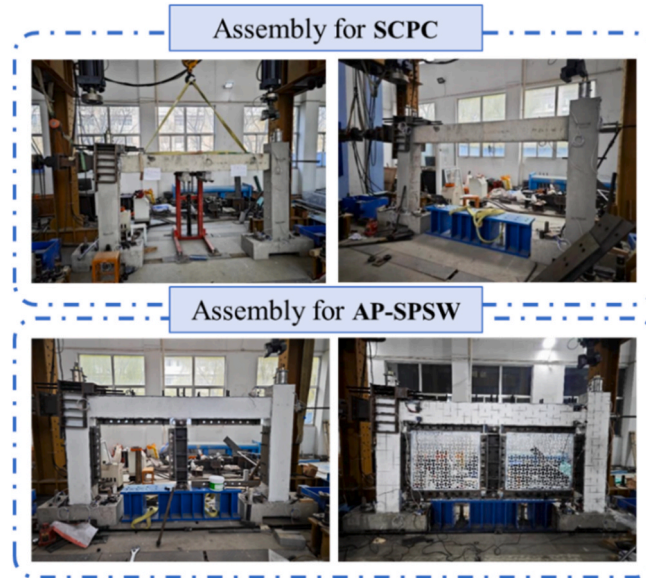


Fig. 8. Assembly for SCPC and AP-SPSW

2.2. Material properties

2.2.1. Concrete

C40 concrete and polypropylene (PP) fiber mortar were used for the prestressed concrete frame structure and joint filling, respectively. According to the concrete material test standard [44], the compressive strengths of the concrete and PP fiber mortar were determined using three standard 150 mm cube specimens. To acquire the actual concrete strength, the testing standard cube specimens were prepared and cured under the same conditions as the test specimens. The average compressive strengths of the concrete and PP fiber mortar over the three cube specimens were determined as 53.25 MPa and 28.02 MPa, respectively.

2.2.2. Steel

The AP-SCPC system utilized various steel materials, including reinforcing bars, prestressing strands, and AP-SPSWs. The infill panels, along with their boundary elements and connection devices, were made from Chinese Q235 steel, while the reinforcing bars consisted of HPB300 and HRB400 grades. To assess the mechanical properties of each steel type, three standard tensile coupon tests were performed. These uniaxial tensile tests adhered to the GB/T228.1–2010 standard [45]. The test results, averaged across all specimens, are summarized in Table 2. The parameters D_s/t , $D_{s,m}/t_m$, $f_{r,y-u}$ and $f_{r,y-l}$, $f_{r,u}$, E_s , and δ in Table 2 represent the design

Table 2

Material properties of steel components.

Steel type	D_s/t (mm)	$D_{s,m}/t_m$ (mm)	$f_{r,y-u}$ (MPa)	$f_{r,y-l}$ (MPa)	$f_{r,u}$ (MPa)	E_s (MPa)	δ
HPB300	6	5.92	381.19	379.13	498.22	2.20×10^5	30.21 %
HRB400	8	8.03	423.19	421.13	607.22	2.02×10^5	28.90 %
	10	9.89	451.25	445.79	622.79	2.02×10^5	28.43 %
	12	11.92	456.28	448.19	637.21	2.02×10^5	27.20 %
	16	16.10	417.91	415.46	601.30	2.02×10^5	25.00 %
	20	20.06	435.99	430.92	629.82	2.02×10^5	20.40 %
Q235	3	2.97	265.29	259.86	363.72	2.06×10^5	28.19 %

diameter/thickness, actual measured diameter/thickness, upper yield strength, lower yield strength, modulus of elasticity, and ultimate elongation, respectively. The prestressing strands used in the test were seven-wire strands with a nominal diameter of 15.2 mm, a yield strength of 1690 MPa, and a tensile strength of 1860 MPa.

2.3. Fabrication and assembly processes

Step 1: The concrete components were fabricated as shown in Fig. 7(a). The reinforcement cages were assembled carefully, and the necessary holes for ducts, bolts, and anchors were reserved in the formwork. Installation of strain gauges on the longitudinal reinforcements was followed by pour of the concrete.

Step 2: The perforated steel plates were cut using laser cutting technology (Fig. 7(b)).

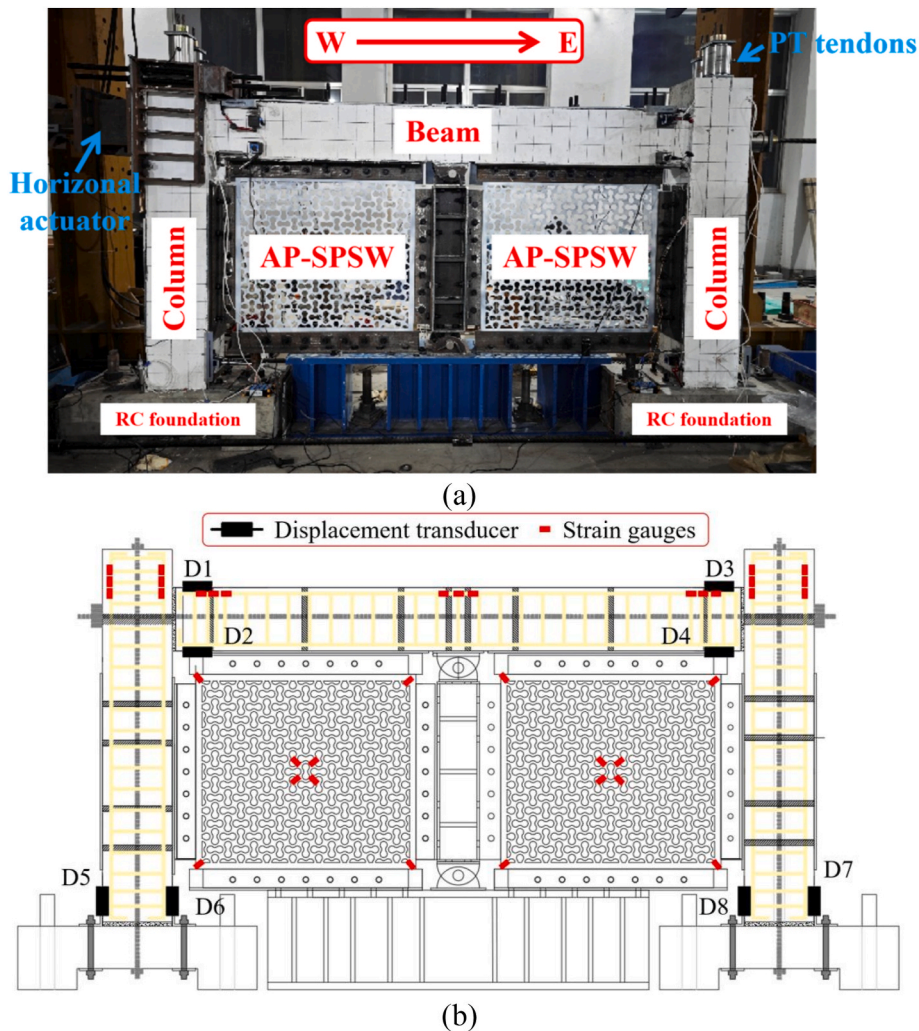


Fig. 9. Test setup, (a) specimen setup, (b) instrumentation.

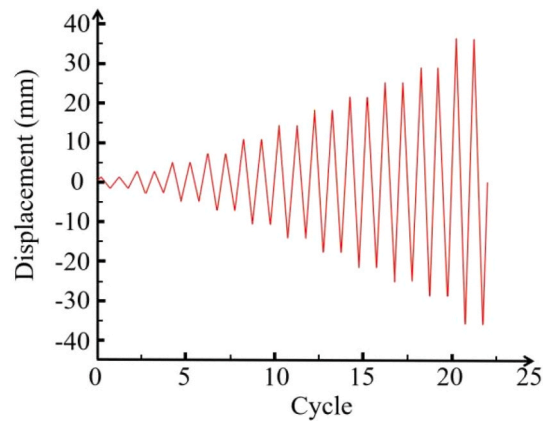


Fig. 10. Loading protocol of the cyclic loading tests.

Step 3: The assemblage process began by positioning the prefabricated columns and RC foundation (Fig. 8). The PT tendons were threaded through the pre-allocated ducts, with anchorages set at the foundation and the tensioning ends aligned with the precast columns. The column-to-foundation connection was then secured to the rigid ground through bolting connections. A column-foundation gap of 20-mm thickness was intentionally left for filling with PP mortar. After the filling process, the mortar underwent a curing period of 72 h to achieve 65 % of its early compressive strength. Prior to tensioning the prestressed steel bars, a pressure sensor was connected to the anchorage to enable real-time monitoring of the PT force during the tensioning process. Tensioning was considered complete when the actual force reached 91 kN, approximately 25 % of the targeted initial PT force (380 kN). As follows, the positioning angle steels at the column-base connection were tightened to the design specifications, thereby completing the assembly of the frame column and foundation.

Step 4: Following the procedure for installing columns, the precast beam was placed with the PT tendons passing through the corresponding ducts. A beam-column gap of 15-mm thickness was similarly filled with mortar before tensioning of the PT strands. Gap filling, mortar curing, and PT strands tensioning were accomplished following the same procedure for the precast columns, with the exception that the PT force reached 50 kN (~25 % of the designed initial PT force (210 kN) in the beam).

Step 5: Before installing the AP-SPSWs, the middle vertical H-shaped hinged device was pre-installed in position (Fig. 8). First, the hinged end caps were installed on the prefabricated columns and steel rigid foundation. Next, the H-shaped steel column was hoisted into position at the hinge joint, and the steel pivot was inserted into the hinge connection holes to complete the installation. The AP-SPSWs were fully-connected to the prefabricated frame using gusset plates: the upper and lower gusset plates were attached to the prefabricated beam and the bottom steel rigid foundation, respectively, and the side gusset plates to both the prefabricated columns and the H-shaped hinged columns. This configuration allows for the use of thinner plates compared to those required for beam-only connections, according to Clayton et al [46]. All connections were fastened using Grade 12.9 high-strength bolts with a diameter of 22 mm. The last procedure following the installation of AP-SPSWs was attributed to tensioning of the PT strands to their targeted initial PT forces (210 kN for beam and 380 kN for columns), after which the entire assembly of the frame specimen was complete.

2.4. Testing procedure

A 1000 kN capacity MTS servo-hydraulic actuator, as illustrated in Fig. 9, was employed to apply loads to the AP-SCPC system. The displacement recording devices consisted of several strain gauges and displacement transducers. Displacement transducers were positioned at each member connection to monitor and record frame lateral displacements, gap openings, and decompression behavior. In total, thirty-seven strain gauges were installed on the AP-SCPC. These included six on the longitudinal reinforcing bars of the eastern and western precast columns, nine on the longitudinal reinforcements at the ends and mid-span of the precast beam, and four located at the corners and center of the two AP-SPSWs. A displacement-dominated loading protocol was adopted during the tests. The designed cyclic displacement levels were determined following ATC-24 [47]. As illustrated in Fig. 10, each peak drift level was repeated for two cycles. As the drift to yielding of infill panels of AP-SPSWs (γ_y) was estimated as 0.35 %, three different levels of peak drifts (~30 %, ~60 %, and 100 % of γ_y) were selected prior to yielding. As follows, eight different levels of peak drifts were adopted. Each loading test was terminated at the end of the second cycle of 2.5 % drift. The peak drifts in sequence were 0.1 % (1.45 mm), 0.2 % (2.90 mm), 0.35 % (5.075 mm), 0.5 % (7.25 mm), 0.75 % (10.875 mm), 1 % (14.5 mm), 1.25 % (18.125 mm), 1.5 % (21.75 mm), 1.75 % (25.375 mm), 2 % (29.0 mm), and 2.5 % (36.25 mm), where the numbers in the brackets denotes the absolute lateral displacements corresponding to the drifts.

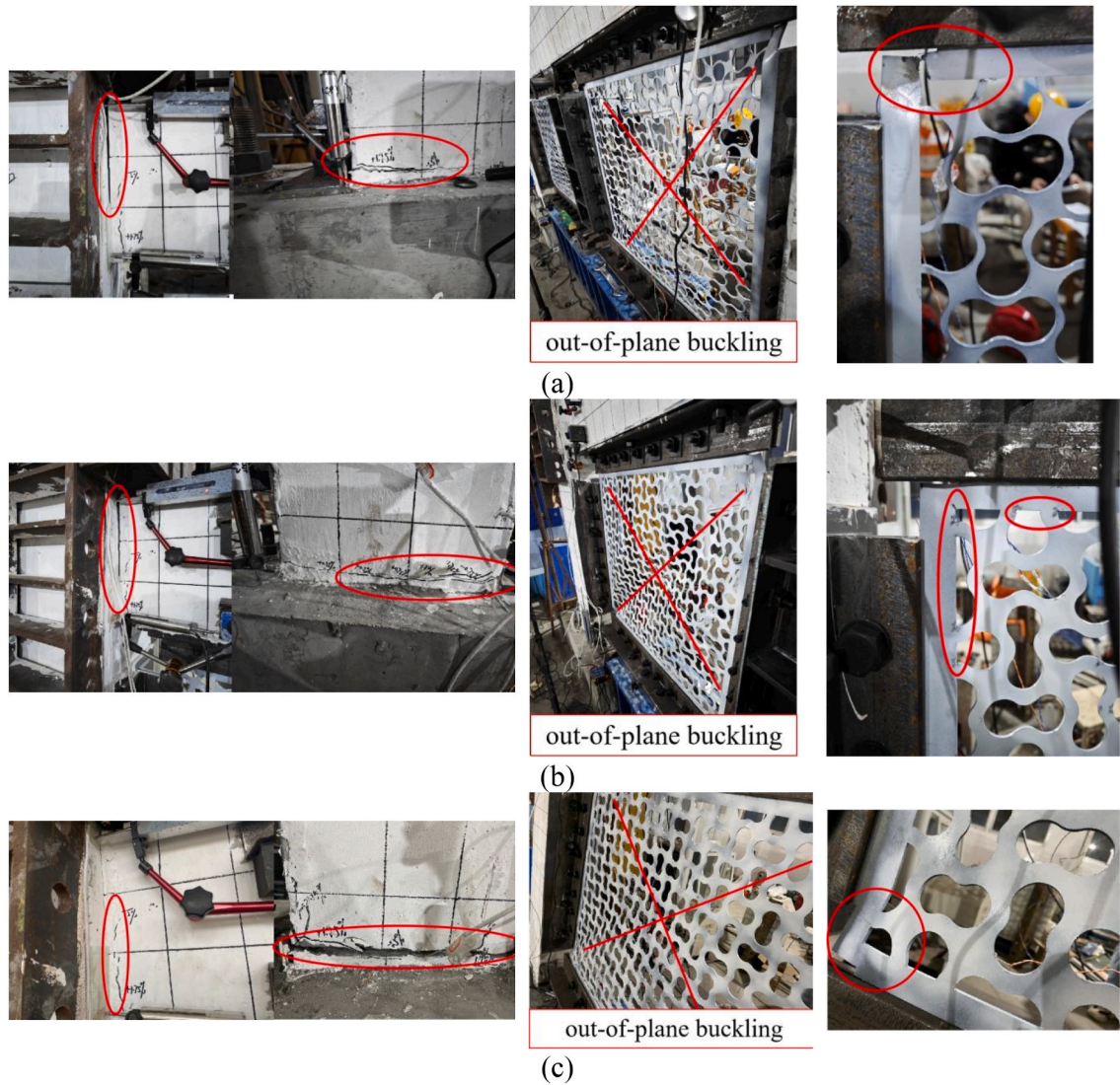


Fig. 11. Failure modes of specimens, (a) OP-0.85-0.4, (b) OP-1-0.4, and (c) OP-1-0.55.

3. Test results and discussed

3.1. Test observations and specimen failure modes

All three specimens demonstrated similar deformation mechanisms and failure patterns. Throughout the tests, the SCPC frames exhibited comparable gap openings and cracking characteristics. Of the three, the OP-0.85-0.4 specimen showed the first signs of slight out-of-plane deformation in the AP-SPSWs at a 0.35 % drift cycle, with the western AP-SPSW typically deforming earlier than the eastern one. At the 1 % drift cycles, all specimens exhibited out-of-plane deformations in their AP-SPSWs. At 1.25 % drift, significant gap opening angles formed at the beam-column joints, with maximum opening displacements around 3 mm and 2 mm on the west side for the OP-1-0.4 and OP-0.85-0.4 specimens, respectively. At the 1.5 % drift cycles, the out-of-plane deformation of the AP-SPSW panels became more pronounced, with X-shaped plastic deformation bands visible in the OP-0.85-0.4 specimen. A fracture appeared at the eastern AP-SPSW's lower-east corner and at the materials connecting the web plate and the boundary elements. For OP-1-0.55, the maximum out-of-plane deformations reached 46 mm on the west side and 31 mm on the east side, increasing to 53 mm and 36 mm at a 1.75 % drift. At the 1.75 % drift cycles, significant increases in gap opening displacements occurred at all the member joints. In OP-1-0.4, a crack developed through the grout at the west column bottom connection, and the maximum gap opening displacement at the east beam-column joint reached 4 mm, accompanied by full-depth grout cracking. At 2 % drift, the infill panels on the OP-1-0.4 specimen experienced maximum out-of-plane deformations of 50 mm on the west side and 45 mm on the east. As the lateral displacement continued to rise, all specimens showed further increases in out-of-plane deformations, along with fractures at the

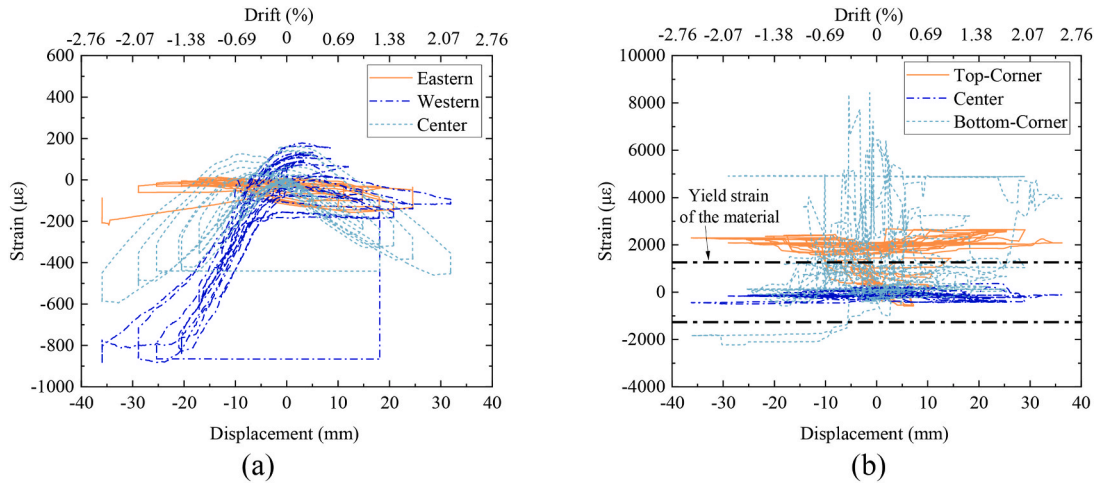


Fig. 12. Strain response of specimen OP-1-0.4, (a) longitudinal reinforcement at beam, and (b) AP-SPSW.

corners between infill panels and connection elements. In OP-1-0.4, the west beam-column joint fully opened, with a maximum displacement of 6 mm at the joint's upper part.

As shown in Fig. 11, the extent of deformation at the four corners and fracture at the ligament connecting the AP-SPSW edges and

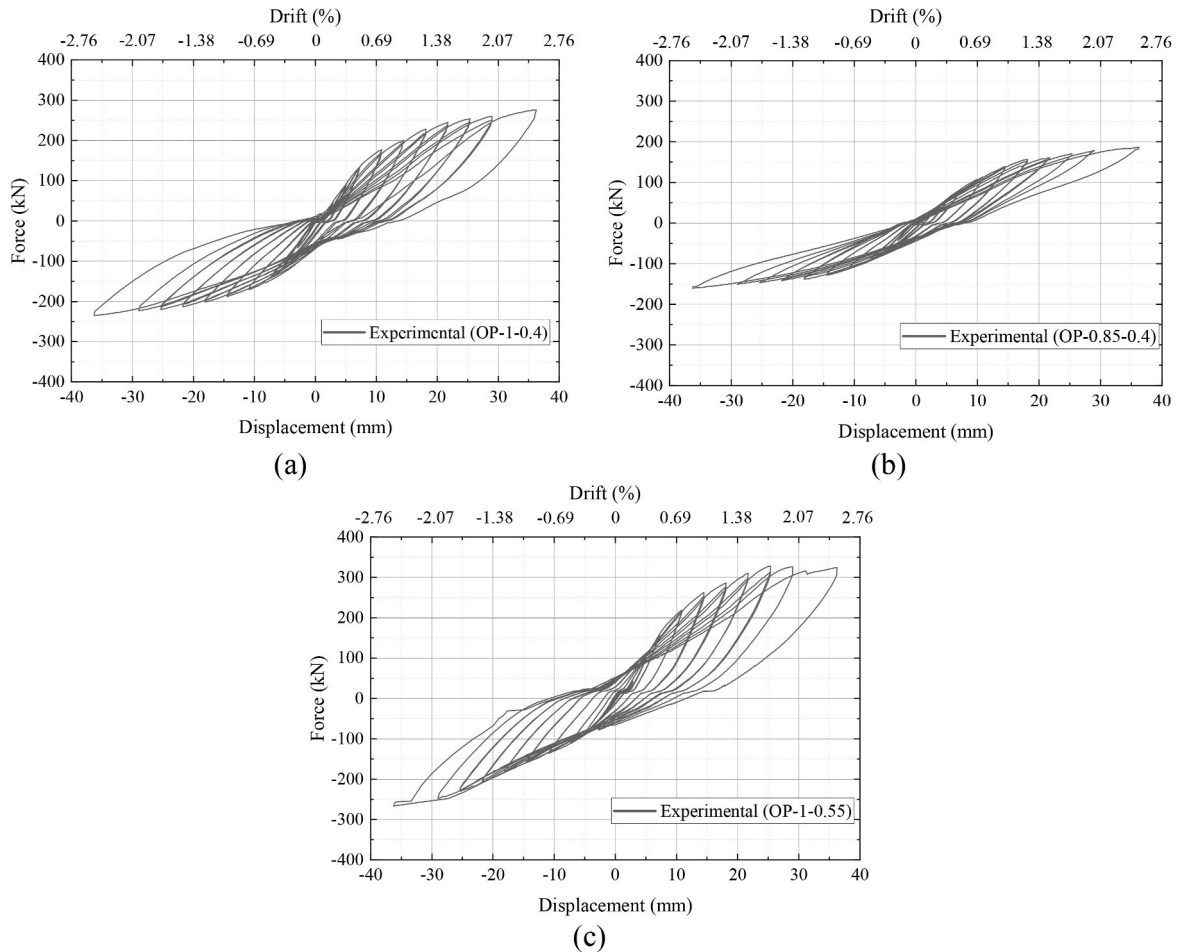


Fig. 13. Hysteresis curve of (a) OP-1-0.4, (b) OP-0.85-0.4 and (c) OP-1-0.55.

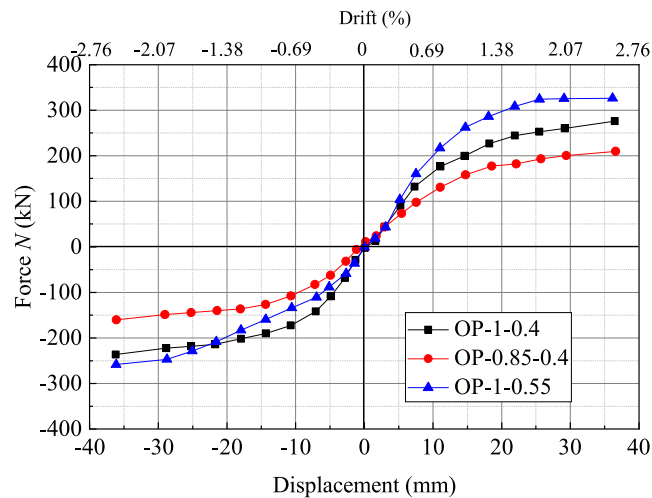


Fig. 14. Skeleton curves of specimens.

infill perforated panel were influenced by the porosity of the AP-SPSW infill plate. Specimens with higher porosity exhibited more pronounced corner deformation. As previously noted, the western AP-SPSW experienced an earlier occurrence and more magnitude of out-of-plane deformation than the eastern side, primarily due to the different lateral displacements imposed to the two AP-SPSWs, as the load was applied only on the western side during testing. The primary failure modes across all three specimens included cracking of the grouted PP fiber mortar, shear buckling of the infill panels, and rupture of the connections between the unit cell ligaments and boundary elements. Notably, the precast RC members remained undamaged, and the connection elements between the beam and AP-SPSWs remained unaffected by the tests.

3.2. Strain results

The strains in the beam's longitudinal reinforcing bars and in the AP-SPSWs were captured via strain gauges. The data for specimen OP-1-0.4 are illustrated in Fig. 12. As seen in Fig. 12(a), the longitudinal reinforcements in the beam showed strains lower than the strain to yielding ($2208 \mu\epsilon$), indicating that these reinforcements remained within the elastic range throughout the test. This observation is consistent with the fact that no visible damage was noted to the beam during the experiment. On the other hand, Fig. 12(b) demonstrates that significant strain occurred at the corner of the AP-SPSW, surpassing the yield strain ($1262 \mu\epsilon$) early at around 0.3 % drift. In contrast, the strain at the center of the AP-SPSW remained within elastic limits, as it did not exceed the yield strain. This implies that plastic deformation could be distributed across a considerable portion of the AP-SPSW's infill panel.

3.3. Hysteresis and skeleton curves

The hysteresis response for the specimens is displayed in Fig. 13, clearly revealing the anticipated flag-shaped behavior. Since the lateral load was applied on only one side, the curves exhibited a leftward shift from the coordinate origin. The pinching observed in the curves suggests that each specimen displayed a reduced residual drift after unloading, attributed to the restoring forces from the PT tendons. However, visible damage to the infill panel at the corners and permanent out-of-plane deformation of the AP-SPSWs impaired the self-centering capability, resulting in incomplete hysteresis loops. Notably, gap openings were detected early in the loading process, around 0.35 % drift, which introduced the specimens with geometric non-linearity.

According to the skeleton curves in Fig. 14, the three specimens behaved elastically prior to the 0.35 % drift cycles. The OP-0.85-0.4 specimen began to appear inelastic response at the first 0.5 % drift cycle. The other two specimens entered into the inelastic stage at likely the 0.75 % drift cycles. The yield point of each specimen was consistent with the moment that the gap opening at each connection and the slight out-of-plane deformation began to appear. Since none of the specimens exhibited significant loss in strength, the ultimate loading carrying capacity was taken as the peak load at the last 2.5 % drift cycle. As seen in Fig. 14, the ultimate loads for specimens OP-1-0.4, OP-0.85-0.4, and OP-1-0.55 were 276.1 kN, 186.5 kN, and 328.3 kN, respectively. It is worth noting that the peak load of the specimens decreased as the porosity of the infill panel of the AP-SPSWs increased. Comparison between OP-0.85-0.4 and OP-1-0.55 indicates that the peak load increased by 76 % as the porosity of unit cell in the AP-SPSWs decreased by 33 %. This is because AP-SPSWs with higher porosity had less material to resist lateral loads, and experienced larger out-of-plane deformation and more serious fracture during loading. The differences in the maximum bearing capacity of the various AP-SCPC specimens are significant, indicating that the SPSW makes a considerable contribution to the bearing capacity of the AP-SCPC. Despite the observation of localized failure of the SPSW during the experiment, the bearing capacity of the AP-SCPC specimens did not exhibit a decline. This can likely be attributed to two factors: 1) the localized failure of the SPSW did not notably reduce its contribution to the bearing capacity of the AP-SCPC, as the SPSW still retained a substantial post-buckling strength; and 2) the contribution of the SCPC bare frame to the

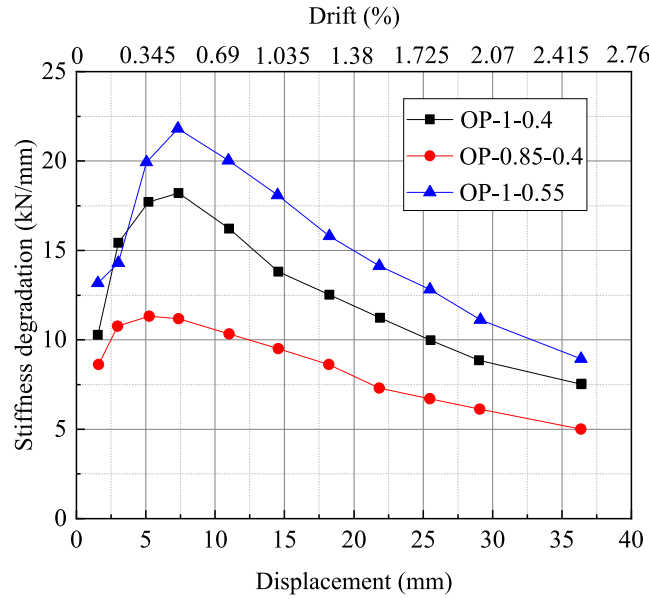


Fig. 15. Stiffness vs. drift curves of specimens.

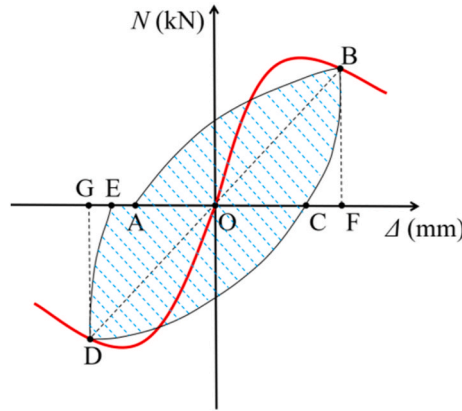


Fig. 16. Illustration of critical load-displacement response quantities in a loading cycle.

bearing capacity of the AP-SCPC increased with the continuous displacement during loading, compensating for the loss of bearing capacity due to the localized failure of the SPSW.

3.4. Stiffness results

The secant stiffness during each cycle, denoted as the i -th cycle, is often employed to characterize the stiffness degradation of a structure or an element subjected to repeated loading. It is determined as follows [9,10,48]:

$$K_i = \frac{(|+F_i| + |-F_i|)}{(|\Delta_i| + |-\Delta_i|)} \quad (3)$$

where $+F_i$ and $-F_i$ are the maximum loads in the positive and negative directions of loading at the i -th drift cycle; $+\Delta_i$ and $-\Delta_i$ are the displacement values corresponding to $+F_i$ and $-F_i$, respectively.

Fig. 15 depicts variation of secant stiffness versus drift for the three specimens. The secant stiffnesses varied in similar manners for the specimens. The initial secant stiffness increased suddenly with the rise in lateral displacement. This is because, as the loading increased, the gaps caused by assembly errors among the components were progressively reduced, resulting in stiffer connections. It is clear that the initial secant stiffness had strong correlation to the porosity of the unit cells of AP-SPSWs. Specimens OP-1-0.55 and OP-0.85-0.4, with porosities of 42.9 % and 63.1 %, exhibited the greatest and smallest initial stiffness, respectively. The peak secant stiffness appeared at 0.5 % drift for OP-1-0.55 and OP-1-0.4 and at 0.35 % drift for OP-0.85-0.4. The drift with peak secant stiffness

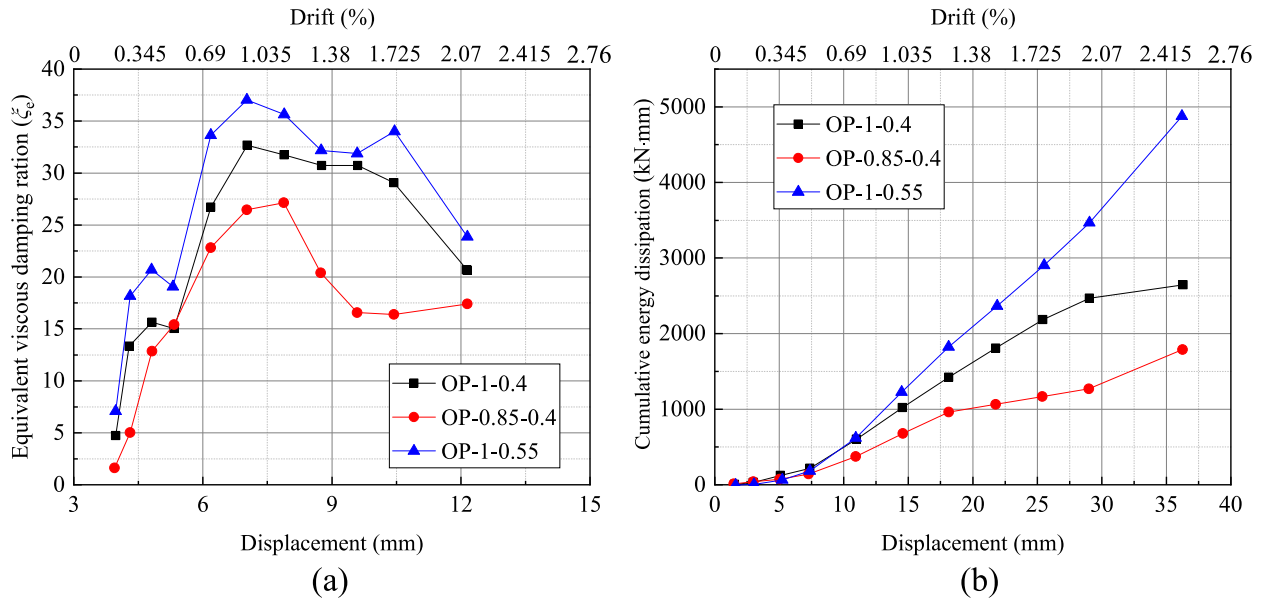


Fig. 17. Energy dissipation, (a) equivalent viscous damping ratio, and (b) cumulative dissipated energy.

corresponded to the moment when visible out-of-buckling deformation presented during the tests. As the lateral displacement continued to increase, the secant stiffness exhibited continuous decrease. This was a result of gap opening in each member joint and the out-of-plane buckling of infill panels of the AP-SPSWs.

3.5. Energy dissipation capacity

Two indicators, i.e., the equivalent viscous damping ratio (ξ_e) and the cumulative dissipated energy (E_{sum}), are commonly used to assess energy dissipation capacity of a structure. The indicators ξ_e and E_{sum} are defined as:

$$\xi_e = \frac{1}{2\pi} \cdot \frac{S_{ABC} + S_{CDE}}{S_{OBF} + S_{ODG}} \quad (4)$$

$$E_{sum} = \sum_{i=1}^n E_i = \int N_i d\Delta_i \quad (5)$$

where S_{ABC} and S_{CDE} denote the areas enclosed by the curves ABC and CDE, respectively, while S_{OBF} and S_{ODG} refer to the areas enclosed by the lines OBF and ODG, respectively (Fig. 16). E_i is the energy dissipated during the i -th cycle, n is the total number of cycles, and N_i and $d\Delta_i$ represent the lateral load and the integral variable for the i -th cycle, respectively.

Fig. 17(a) and (b) illustrate the equivalent viscous damping ratio and cumulative dissipated energy of each specimen, respectively. The curves for ξ_e display a similar trend, rising twice before declining. The dual rise can be attributed to the sequential energy dissipation contributions from the SCPC frames and the AP-SPSWs. After the 1.25 % drift cycle, the curves declined due to the onset of plastic deformation in the AP-SPSWs. As is depicted in Fig. 17(b), all specimens exhibited minimal cumulative energy dissipation (E_{sum}) prior to 0.35 % drift. However, as drift increases, significant plastic deformation of the infill panels resulted from out-of-plane buckling led to enhanced energy dissipation. Notably, the dissipated energy of the AP-SCPC increased as the porosity of the AP-SPSWs decreased. This is because more content of steel in the infill panel may participate in the plastic deformation as the porosity decreased. However, as suggested by Wang et al. [40], the auxetic effect and deformation capacity of the AP-SPSWs depends on the rotational movement among the unit cells. A decrease in porosity negatively affects the rotational capacity of each unit cell, thereby reducing the yielding of the infill panels AP-SPSW, which is unfavorable for energy dissipating of the AP-SPSW specimen.

3.6. PT tendon behavior

The PT force was recorded using pressure sensors placed near the anchorages. Fig. 18 illustrates the variation of the PT forces with drift in OP-1-0.4. The curves show asymmetry because the applied load was only on the western column. Prior to the opening of the gaps, PT force losses were relatively small, amounting to 3.52 %, 8.06 %, and 1.56 % of the initial magnitudes in the beam, western, and eastern column, respectively. Anchorage retraction was mainly responsible for this reduction in prestress. As the gap opening occurred at the joint interfaces, the PT tendons stretched elastically, causing a linear rise in PT force. At the first 2.5 % drift cycle, the peak PT forces reached 452 kN in the western column, 447 kN in the eastern column, and 285 kN in the beam, demonstrating that the

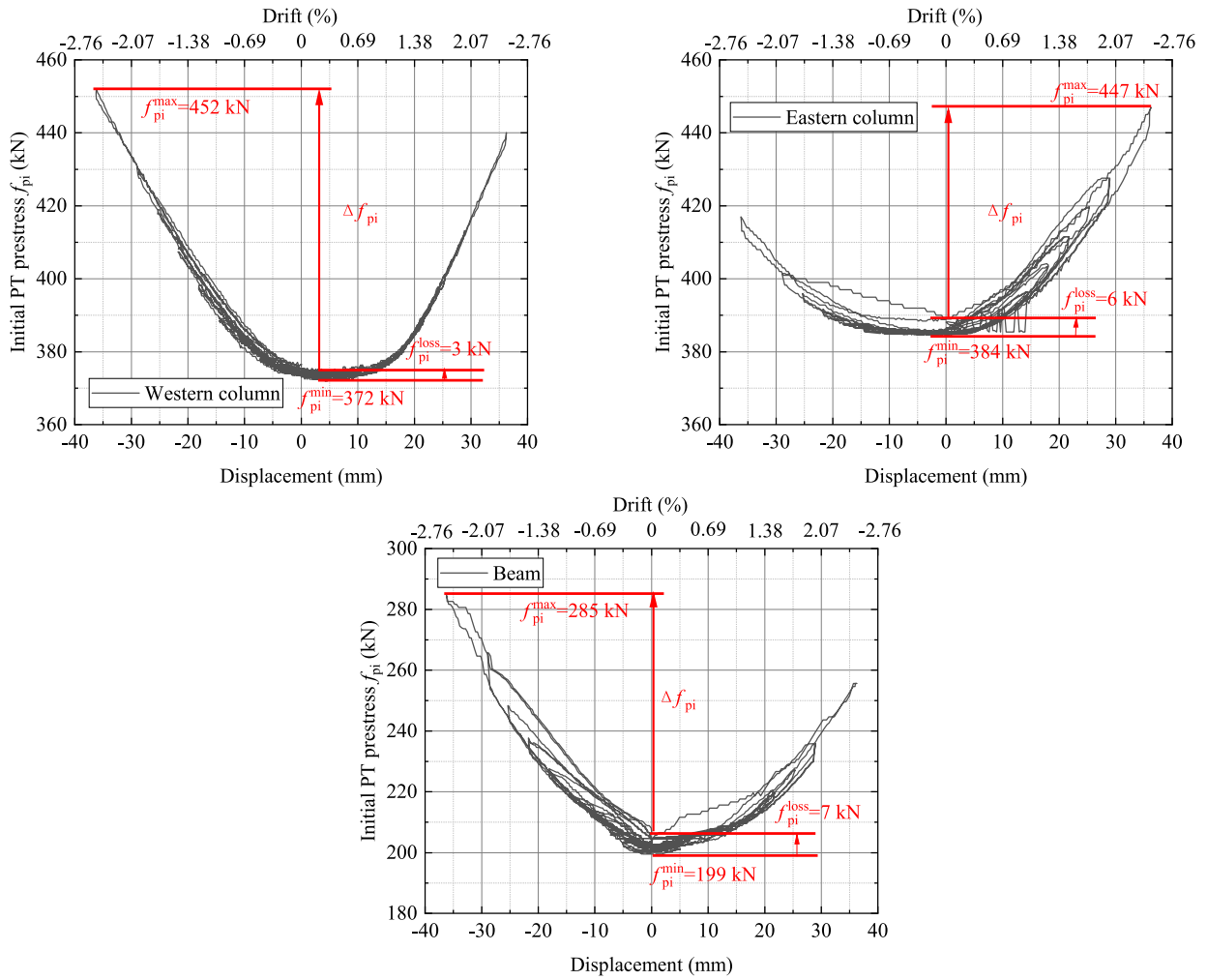


Fig. 18. PT forces of OP-1-0.4.

PT tendons remained within their elastic range.

Based on these observations, the PT force (f_{PT}) can be computed using the linear equation given as:

$$f_{PT} = f_{PT,i} + k_{pt} (\delta_{pt} - \delta_s) \quad (6)$$

where $f_{PT,i}$ is its initial value, $k_{pt} = A_s E_s / L_s$, and δ_{pt} and δ_s represent the axial stiffness and the PT elongation owing to gap opening, and the shortening of the RC member, respectively. The parameters L_s , E_s , and A_s denote the unbonded length of the PT strands, the area of cross-section, and the modulus of elasticity, respectively. The gap opening displacement at the location of PT strands can be given as:

$$\delta_{pt} = \theta_r \left(\frac{h}{2} - c \right) \quad (7)$$

where h is the sectional depth of a RC member, and the gap-opening angle θ_r can be expressed as:

$$\theta_r = \frac{|g_1| + |g_2|}{h} \quad (8)$$

and c is the distance from the neutral axis to the extreme compression fiber in a member, which is given as:

$$c = \frac{|g_2|}{|g_1| + |g_2|} h \quad (9)$$

where g_1 and g_2 denote the gap-opening and decompression displacement at the extreme fibers, respectively. The axial shortening of a RC member can be calculated based on the total axial compression (N) applied to the member:

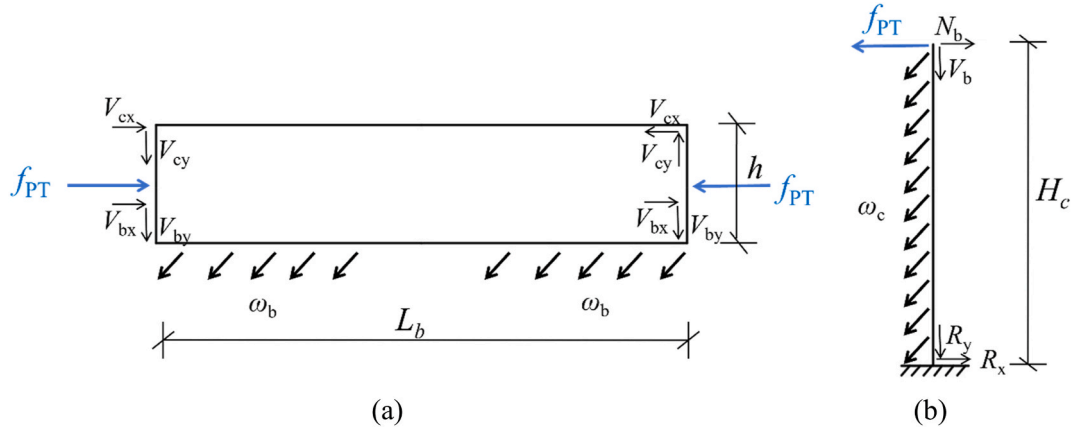


Fig. 19. Free-body diagram of precast members, (a) beam, and (b) column.

$$\delta_s = \frac{N}{k_s + k_{pt}} \quad (10)$$

where k_s is the axial stiffness of the reinforced concrete (RC) member.

To illustrate the derivation of axial compression in members, this section uses the precast beam as an example. Referring to the free-body diagram in Fig. 19, the total axial force in the beam is contributed from: 1) the PT force (f_{PT}), 2) the horizontal component of the distributed forces acting from the AP-SPSWs, and 3) the end reaction forces owing to the column shear forces ($P_{b(column)}$). The beam axial load (N_b) can be thus expressed as:

$$N_b(x) = f_{PT} + P_{b(column)} + \frac{1}{2}P_{b(web)} - \omega_{bx}x \quad (11)$$

where $P_{b(web)}$ is the end reaction force due to the distributed forces from the AP-SPSWs, x is the coordinate along the beam length. It is important to note that the axial force in the beam varies linearly along its length due to the distributed loads from the AP-SPSWs. The reaction forces at the beam ends $P_{b(web)}$ and $P_{b(column)}$ can be expressed as:

$$P_{b(web)} = \omega_{bx}L_b \quad (12)$$

$$P_{b(column)} = \frac{1}{2}\omega_{cx}H_c \quad (13)$$

where L_b and H_c are the beam length and column height, respectively, and ω_{bx} and ω_{cx} are the horizontal components of the distributed loads applied to the beam and column from the web panel, respectively, which can be computed as:

$$\begin{aligned} \omega_{bx} &= \frac{V_b^{SPSW}}{L_b} \\ \omega_{cx} &= \frac{V_c^{SPSW} \tan \alpha}{H_c} \end{aligned} \quad (14)$$

where V_b^{SPSW} and V_c^{SPSW} represent the shear forces acting on the beam and column, respectively, and α is the angle of the inclined tension strips in the AP-SPSWs after yielding, measured relative to the vertical. The shear forces from the AP-SPSWs can be computed using the method proposed by Wang et al. [40]:

$$\begin{aligned} V_b^{SPSW} &= 0.42Rf_y t_w L_b \\ V_c^{SPSW} &= 0.42Rf_y t_w H_c \end{aligned} \quad (15)$$

where R is a modification factor accounting for the effects of auxetic-shaped perforations, as recommended by Wang et al. [40]:

$$R = 0.56 \frac{N_r L_0 (1 - \Phi)}{2H_w} \quad (16)$$

where N_r is the number of unit cells included along the length of the diagonal strip, and H_w is the height of the infill panel of the AP-SPSWs.

With the explicit expression of beam axial force, the total beam axial shortening due to compression load can be calculated as:

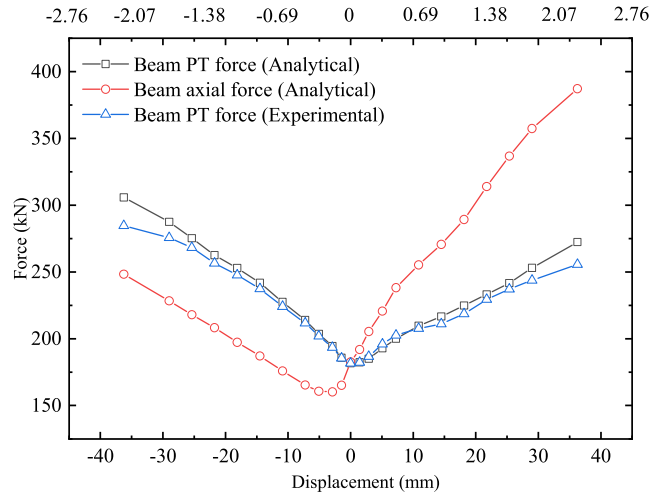


Fig. 20. Comparison of the measured and theoretical PT force in the beam of OP-1-0.4.

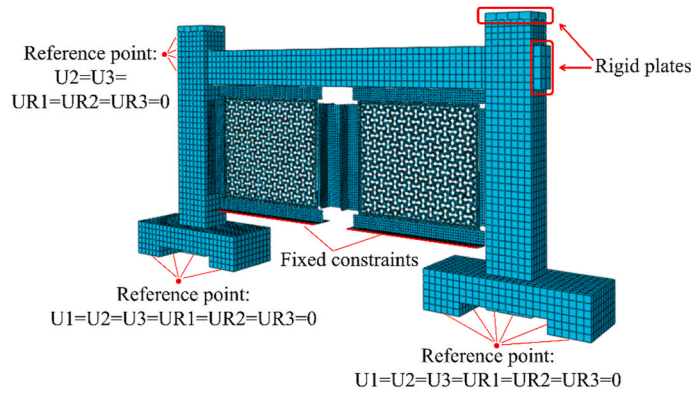


Fig. 21. Illustration of FE model.

$$\delta_b = \int_{L_b} \frac{N_b(x)dx}{(k_b + k_{pt})L_b} \quad (17)$$

where k_b is the axial stiffness of beam and can be taken as $A_b E_c / L_b$, where A_b and E_c are the cross-sectional area and equivalent elastic modulus, respectively.

Fig. 20 compares the measured and analytical PT forces to assess the proposed equations. The analytical solutions show good agreement with the measured PT forces during the initial loading stage. However, as lateral displacement increases, the analytical values tend to exceed the test results. Two factors contribute to this discrepancy: (i) the theoretical model assumes that the PT tendons remain centered within the ducts, thereby neglecting their contribution to the column shear forces, and (ii) the analytical equations are based on the simplified assumption that the web plates are fully yielded, which may overestimate the forces in the web plates, leading to unconservative results.

4. Finite element analysis

4.1. Development of the numerical model

A numerical model of AP-SCPC subassembly was developed using ABAQUS platform [49] to acquire more detailed insights into the seismic behavior of system. A schematic view of the model is presented in Fig. 21, which consists of a precast beam, precast columns, RC foundation, AP-SPSWs, PT tendons, angle steels, and high-strength bolts. In the FE model, concrete components and steel reinforcements were discretized using 8-node 3-dimensional solid elements with reduced integration point (C3D8R) and linear truss elements (T3D2), respectively. The AP-SPSWs along with the connection devices were discretized using 4-node shell elements (S4R). To ensure computational efficiency and convergence without compromising result accuracy, the mesh sizes for concrete and steel

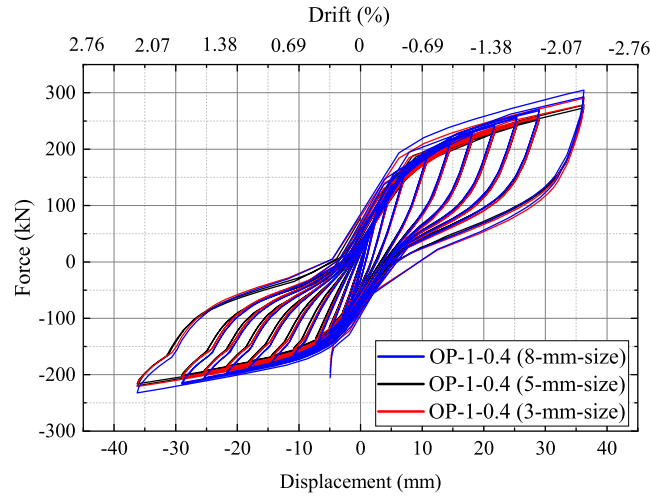


Fig. 22. Results of OP-1-0.4 from mesh sensitivity studies on the AP-SPSW component.

Table 3

Values of parameters used in CPD model.

e	ψ	ν	K_c	f_{bo}/f_c	f_c (MPa)
0.1	30°	0.00005	0.6667	1.16	53.25

components were selected as 50 mm and 20 mm, respectively, based on mesh dependency studies. Mesh sensitivity studies were conducted to investigate the impact of element size of AP-SPSW component on the AP-SCPC model, using three different mesh sizes (8 mm, 5 mm, and 3 mm). The results for Specimen OP-1-0.4 are presented in Fig. 22. It is evident that the models with moderate and fine meshes (5 mm and 3 mm) exhibited almost identical hysteretic responses. Considering both computational efficiency and accuracy, a moderate mesh size of 5-mm elements was selected for the AP-SPSW model.

Tie constraints were applied between two contacted surfaces of the anchorage and precast columns. Similarly, tie constraints were used for the adjacent surfaces in the AP-SPSWs and their connection devices. Surface-to-surface contact interaction was assigned at the interfaces between each two precast components. This contact interaction included a “hard contact” and a penalty with friction coefficient of 0.6 in the normal and tangential directions, respectively [50]. The embedded constraints available in ABAQUS were effectively utilized to replicate the behavior of steel reinforcements within a concrete matrix. To model unbonded interactions between the PT tendons and concrete, there is no interaction defined. Instead, local coordinate systems were established and truss elements were created along the length of PT tendons. Springs constraints with a stiffness of 1×10^5 N/mm were used to link the truss elements to the PT tendons to ensure rigid-body behavior and thus the truss elements deformed consistently with the PT tendons. Similarly, embedded constraints were used to simulated interactions the truss elements and concrete. Three reference points were established neighboring to the beam prestressing anchorages and below the bottom of the RC foundations, which were coupled with their respective entire side surfaces. The bottom ends of the RC foundations were fully fixed. The reference point adjacent to the left beam PT

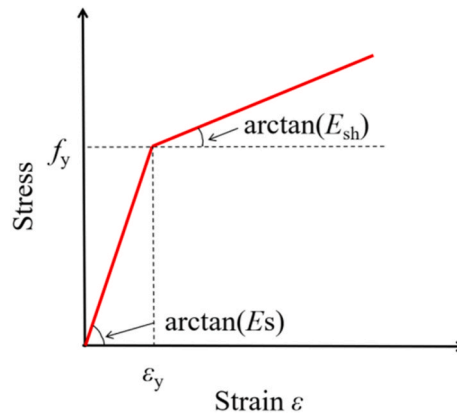


Fig. 23. Material model for reinforcements and steel components.

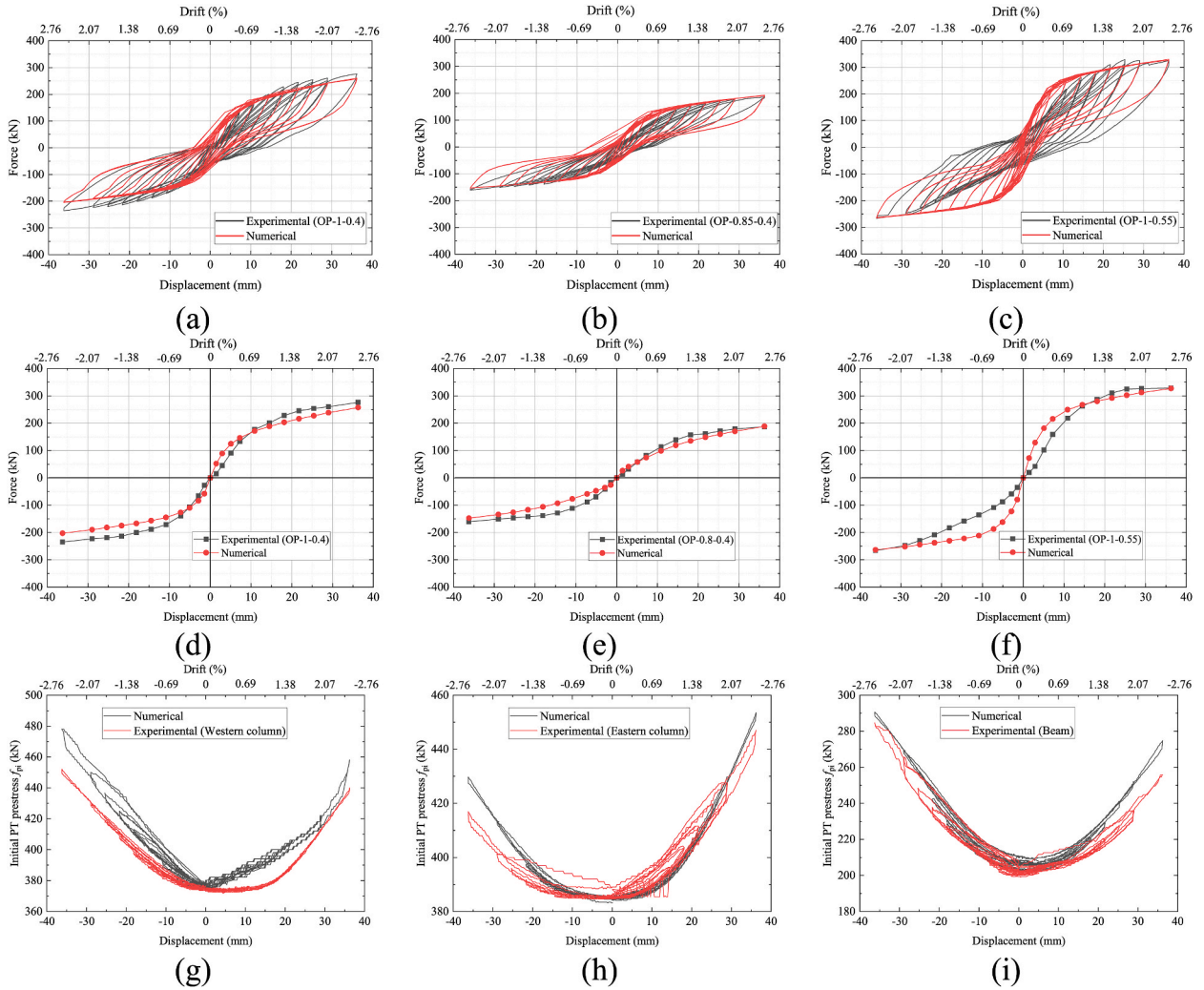


Fig. 24. Comparison of test and numerical results, (a)–(c) hysteresis response, (d)–(f) skeleton load-displacement response, and (g)–(i) PT forces of OP-1-0.4.

anchorage was constrained in its all except the horizontal translational degree of freedom. To mimic the actual boundary conditions at the bottom of the AP-SPSWs while improve computational efficiency, the bottom surfaces of HBES of the AP-SPSWs were with fixed constraints. The lateral cyclic loads were directly applied to the top reference point. All the boundary conditions applied to the specimen are summarized in Fig. 21. Initial prestress of tendons was generated through a predefined temperature field using the following relationship:

$$\sigma_p = C\alpha E\Delta T \quad (18)$$

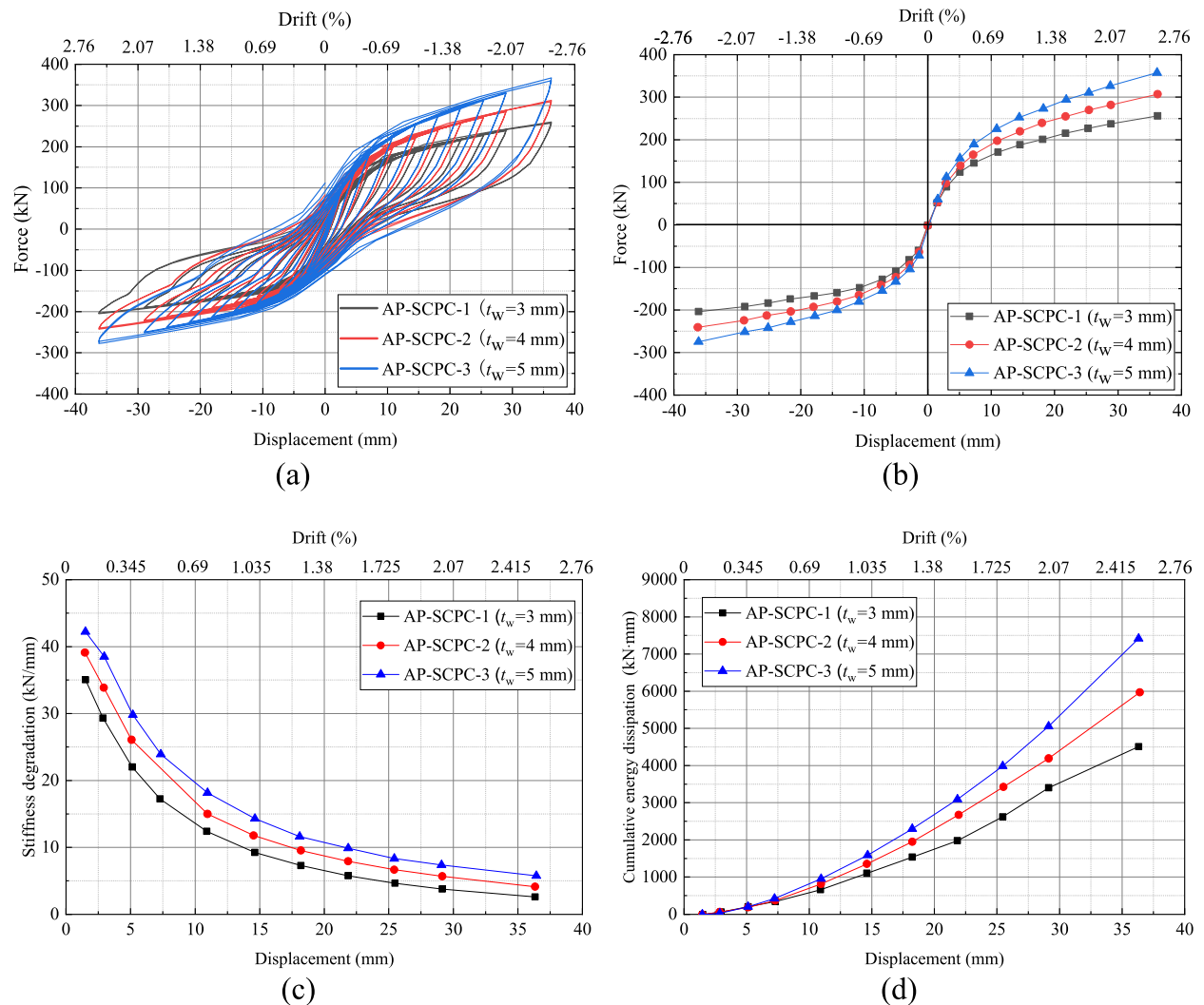
where σ_p is the initial prestress, C is the correction factor, α is the coefficient of linear expansion, E is modulus of elasticity and ΔT is the temperature change.

Concrete damaged plasticity (CDP) model available in ABAQUS was employed to represent the elastoplastic behavior of concrete under compression and tension. The parameters for the CDP model include flow potential eccentricity (e), dilation angle (Ψ), viscosity parameter (ν), ratio of the second stress invariant on the tensile meridian to that on the compressive meridian (K_c), and ratio of the strength under biaxial compression to that under uniaxial compression (f_{bo}/f_c). The values of parameters used in the modeling are listed in Table 3. A bilinear model was used to describe the constitutive behavior of the reinforcements and steel components, as shown in Fig. 23, where the strain hardening modulus (E_{sh}) was calculated as $(f_{r,u} - f_{r,y})/(\epsilon_u - \epsilon_y)$ and $0.02E_s$, respectively. The quantities $f_{r,y}$ and $f_{r,u}$ are the yield and ultimate strengths of the reinforcements, and ϵ_y and ϵ_u are the corresponded strains. The combined isotropic-kinematic cyclic hardening was employed to represent the steel plasticity under cyclic loading [35].

Table 4

Values of parameters and results of the parametric studies of AP-SCPCs.

Specimens	t_w (mm)	r_1 (mm)	r_2 (mm)	D (mm)	r_1/r_2	r_1/D	f_{pi}/f_{ptk}	E_{last} (kN-mm)	E_{sum} (kN-mm)	ξ_e	Δ_{last} (mm)
AP-SCPC-1	3	17	17	43	1.00	0.40	0.35	4836.00	8608.20	16.03	4.14
AP-SCPC-2	4		17	43	1.00	0.40	0.35	6166.65	10418.87	17.32	8.37
AP-SCPC-3	5		17	43	1.00	0.40	0.35	7803.29	12843.21	18.68	11.76
AP-SCPC-4	3		20	43	0.85	0.40	0.35	3447.45	6485.48	15.71	3.89
AP-SCPC-5	3		14	43	1.20	0.40	0.35	5391.89	8427.37	22.33	18.26
AP-SCPC-6	3		17	57	1.00	0.30	0.35	3548.51	6713.71	14.09	4.91
AP-SCPC-7	3		17	31	1.00	0.55	0.35	5489.85	10274.74	14.62	3.49
AP-SCPC-8	3		17	43	1.00	0.40	0.40	5127.40	9498.30	16.79	5.38
AP-SCPC-9	3		17	43	1.00	0.40	0.30	4566.13	7899.95	15.32	5.97

**Fig. 25.** Effect of web thickness of AP-SPSWs on the hysteresis response of AP-SCPCs, (a) hysteresis response, (b) skeleton load-displacement response, (c) secant stiffness, and (d) dissipated energy.

4.2. Model validation

Test results are employed as the benchmark for model validation through comparison of (i) the hysteresis curve, (ii) the skeleton curve, and (iii) the column PT forces for OP-1-0.4. The comparable results of hysteresis response shown in Fig. 24(a)–(c) suggest that the numerical model provides a good prediction of hysteresis response. The numerical curves exhibit a more pronounced “pinching” phenomenon. It seems that the numerical model underestimates the permanent drifts due to mortar damage and crushing. In addition,

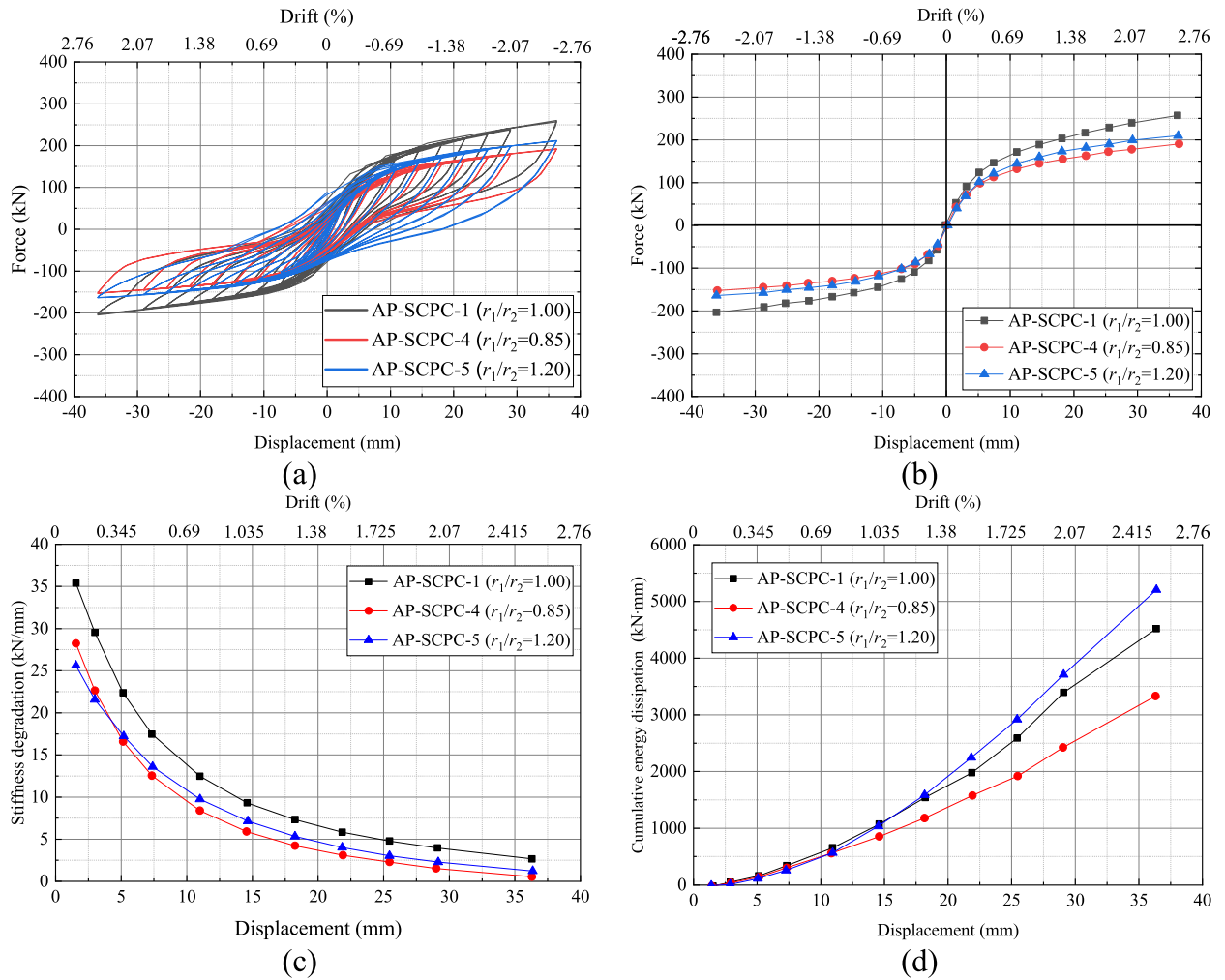


Fig. 26. Effect of r_1/r_2 of AP-SPSWs on the hysteresis response of AP-SCPCs, (a) hysteresis response, (b) skeleton load-displacement response, (c) secant stiffness, and (d) dissipated energy.

the calculated load-carrying capacity at each given drift level by the numerical model was with very high accuracy (Fig. 24(d)–(f)). Fig. 24(g)–(i) compares the PT forces for OP-1-0.4 obtaining from modeling and testing. The numerical results likely overrate the PT forces, primarily because the anchorage shrinkage induced loss of PT force is not considered in the model. Nevertheless, the comparisons validate the numerical model, suggesting its reliability for use of parametric studies of the effects of critical design parameters.

5. Parametric studies

Based on the previously developed FE model, the hysteresis behavior of AP-SCPCs was thoroughly investigated in this section. A total of nine specimens were simulated, with variables including the thickness of the AP-SPSWs (t_w), the ratio of the head to side circle radii (r_1/r_2), the ratio of the radius to the centroid distance of the two head circles (r_1/D), and the initial PT prestress (f_{pi}/f_{ptk}). The details of the parameters and the results obtained from the FE models are presented in Table 4, where E_{last} and Δ_{last} represent the dissipated energy for the last cycle and the corresponding residual deformations, respectively.

5.1. Effect of thickness of AP-SPSWs (t_w)

The influence of varying AP-SPSW infill panel thicknesses (t_w) on the hysteresis response of AP-SCPCs was investigated by adopting three different thicknesses, i.e., 3, 4, and 5 mm. As shown in Fig. 25(a) and (b), the peak force and post-yielding stiffness of the specimens increased with increasing t_w . It is also indicated that thicker infill panels lead to more pronounced residual deformations, which indicate a weaker self-centering capacity of the specimen. Therefore, selecting an appropriate thickness for AP-SPSWs is crucial to achieving a favorable balance between load-carrying capacity and self-centering ability. Comparing SPSW-1 with SPSW-2 (or SPSW-3), it is evident that increasing the infill panel thickness from 3 mm to 4 mm (or 5 mm) raised the peak force from 256.97 kN to 309.03

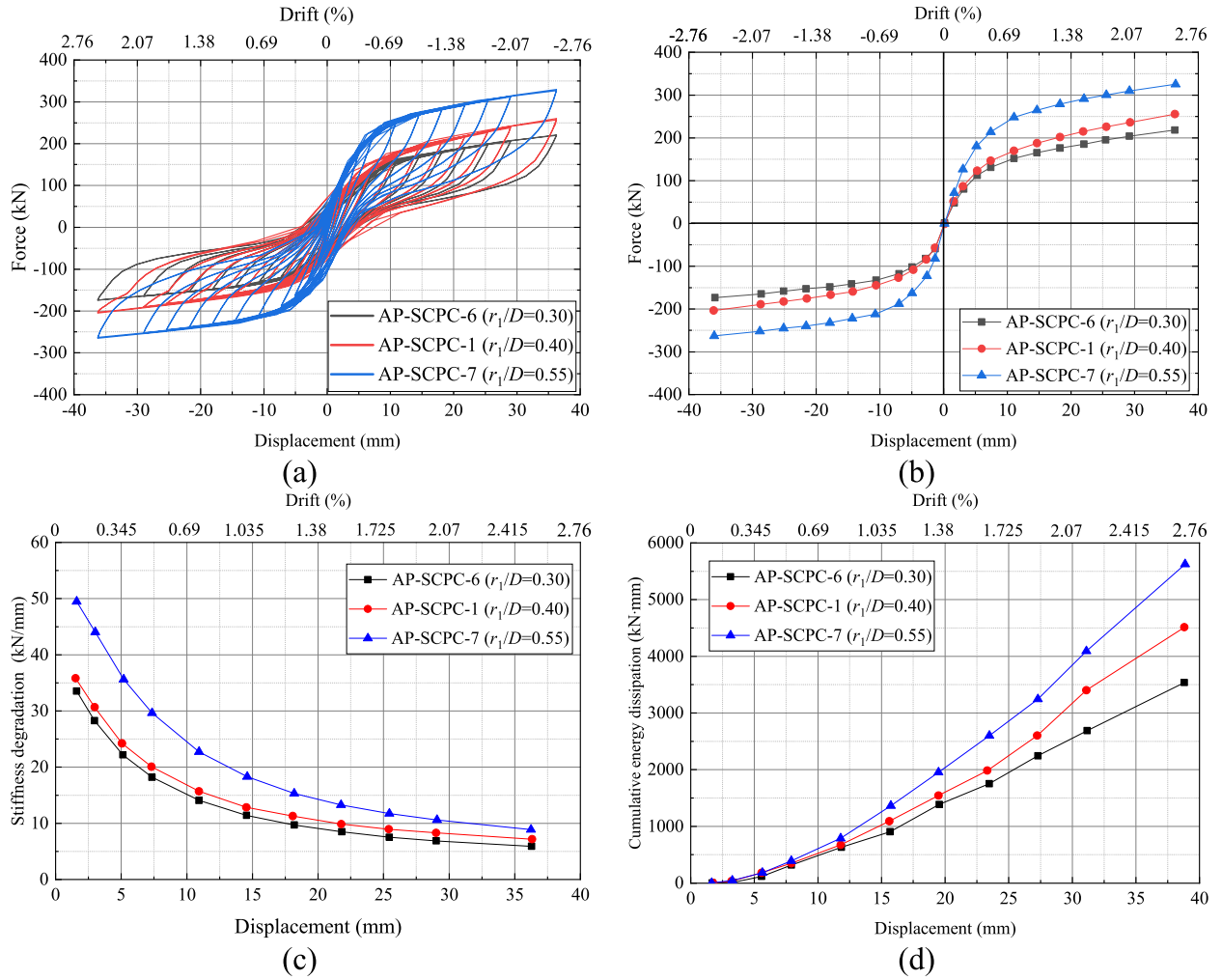


Fig. 27. Effect of r_1/D of AP-SPSWs on the hysteresis response of AP-SCPCs, (a) hysteresis response, (b) skeleton load-displacement response, (c) secant stiffness, and (d) dissipated energy.

kN (or 399.6 kN), representing an increase of 30 % (or 43 %). A similar trend was observed for secant stiffness, as shown in Fig. 25(c). Fig. 25(d) depicts that the specimens' dissipated energy capacity also increased with t_w . According to Table 4, the dissipated energy of SPSW-1 increased from 4836.00 kN·mm to 6166.65 kN·mm for SPSW-2 (or 7803.29 kN·mm for SPSW-3), representing an increase of 28 % (or 61 %).

5.2. Effect of r_1/r_2

The effect of the r_1/r_2 ratio on specimen hysteresis response was evaluated using three values: 0.85, 1.00, and 1.20. The r_1/r_2 ratio significantly impacts the specimens' peak force, stiffness, and dissipated energy capacity (Fig. 26). This is because changes in r_1/r_2 alter the distance between neighboring perforations, affecting the auxetic effect and deformation capacity of the AP-SPSWs, which subsequently influences the hysteresis performance of SCPCs. Comparing SPSW-4 with SPSW-1 (or SPSW-5), it is evident that the dissipated energy increased from 3447.45 kN·mm to 4836.00 kN·mm (or 5391.89 kN·mm), representing a 40 % (or 56 %) increase. Additionally, residual deformations increased with r_1/r_2 . The AP-SPSW with $r_1/r_2 = 1$ provided the greatest load-carrying capacity and stiffness, with moderate energy dissipation, aligning with experimental results. Therefore, an r_1/r_2 ratio of 1 is recommended for the design of AP-SPSWs.

5.3. Effect of r_1/D

Three different magnitudes of the aspect ratio of the peanut-shaped cutout, i.e., $r_1/D = 0.30, 0.40$, and 0.55 , were employed to evaluate the effect of this parameter. The results from the FE models are presented in Fig. 27 and Table 4. The peak load, initial stiffness, dissipated energy capacity, and secant stiffness all increased with a higher r_1/D ratio. Comparing SPSW-6 with SPSW-1 (or

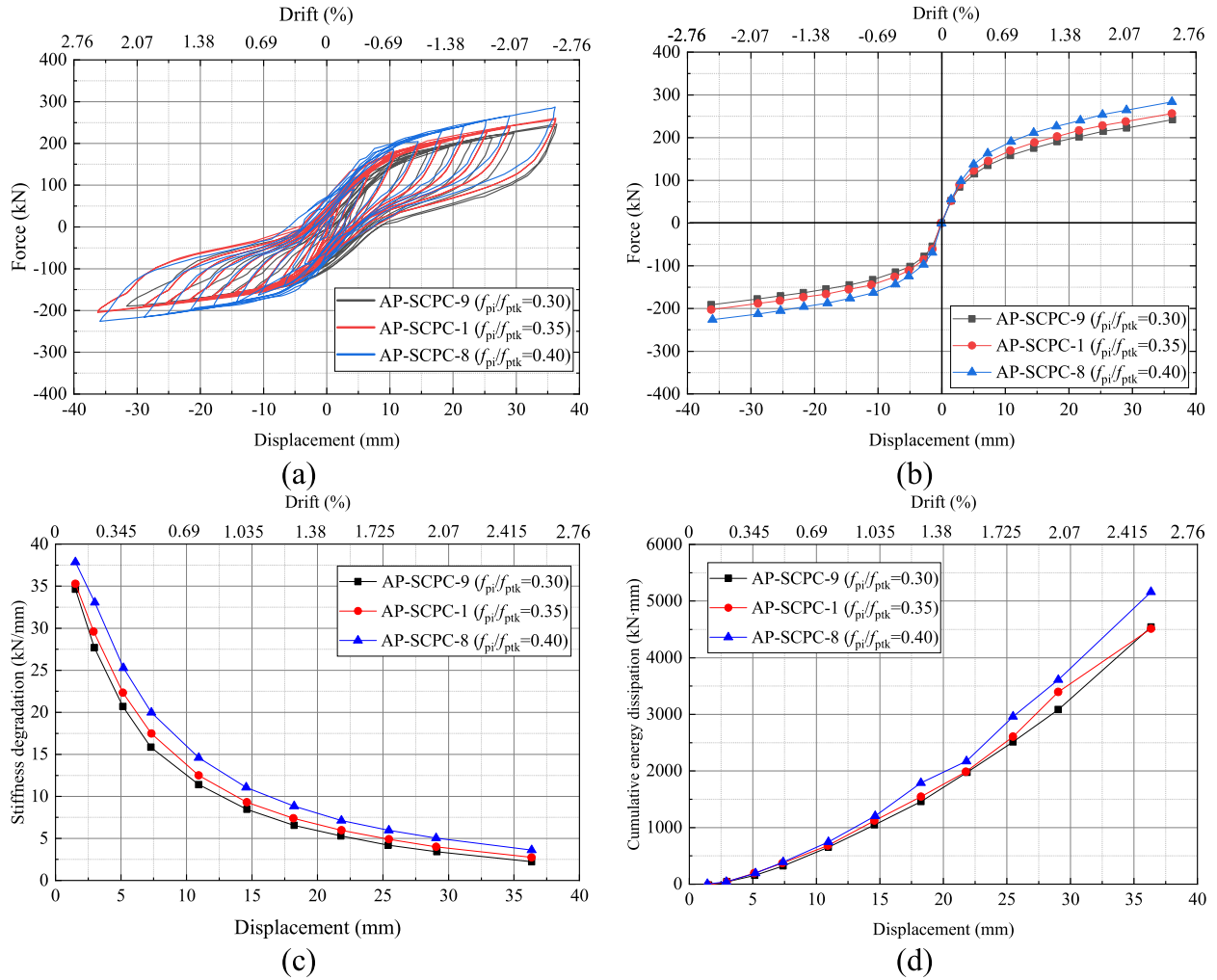


Fig. 28. Effect of initial PT prestress (f_{pi}/f_{ptk}) on the hysteresis response of AP-SCPCs, (a) hysteresis response, (b) skeleton load-displacement response, (c) secant stiffness, and (d) dissipated energy.

SPSW-7), it is evident that the dissipated energy capacity increased from 3548.51 kN•mm to 4836.00 kN•mm (or 5489.85 kN•mm), representing an increase of 36 % (or 55 %). The specimens exhibited similar residual deformations. Therefore, increasing the r_1/D ratio is likely an effective way to improve the hysteresis performance of AP-SCPCs.

5.4. Effect of initial PT prestress (f_{pi}/f_{ptk})

The effect of initial PT prestress was evaluated by varying its value twice, i.e., $f_{pi}/f_{ptk} = 0.3, 0.35$, and 0.4 , where f_{pi} and f_{ptk} represent the initial prestress and ultimate tensile strength of PT tendons, respectively. The hysteresis curves shown in Fig. 28(a) indicate that the “pinching” phenomenon becomes more pronounced with higher initial PT prestress. Increasing the initial PT force improves the self-centering capacity of the structural system by reducing its residual deformation. Fig. 28(b) and (c) shows that both the loading-carrying capacity and the secant stiffness of the AP-SCPC increase with the initial PT prestress. Fig. 28(d) demonstrates that changing the initial PT prestress has little effect on the dissipated energy capacity of the AP-SCPC.

5.5. Comparison of AP-SCPC and SP-SCPC

To evaluate the impact of SPSW perforations on the cyclic behavior of AP-SCPCs, two additional SCPC models with conventional (non-perforated) SPSWs were developed. These models featured different infill panel thicknesses: one with $t_w = 3.00$ mm, identical to specimen OP-1-0.4, and the other with $t_w = 1.57$ mm, designed to maintain the same steel content as OP-1-0.4. The hysteresis curves of the conventional SCPC models, presented in Fig. 29(a), reveal asymmetry between the positive and negative loading directions. Specifically, the load-carrying capacity in the negative loading direction is significantly lower than in the positive direction. As shown

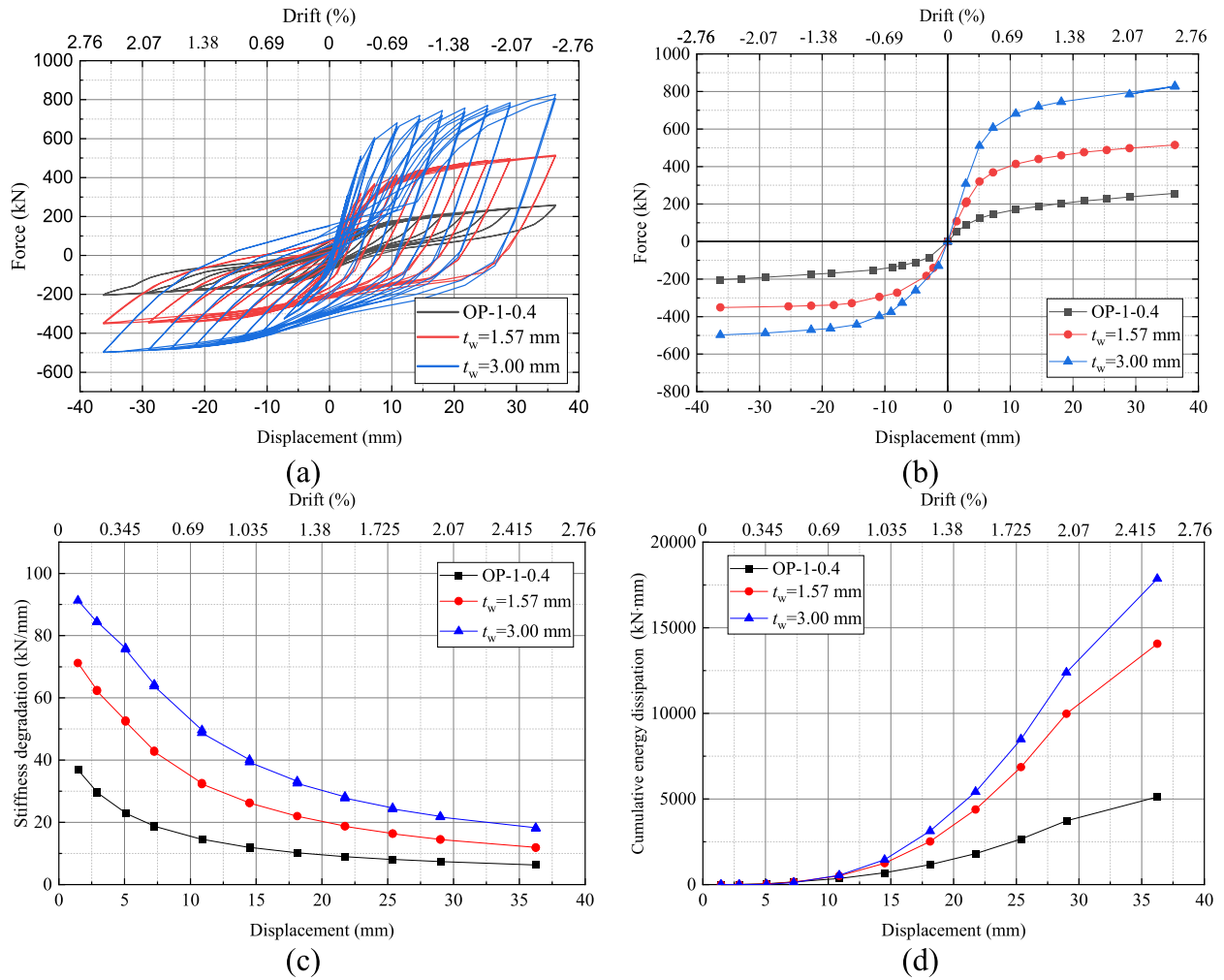


Fig. 29. Effect of perforation of infill panels on the hysteresis response of AP-SCPCs, (a) hysteresis response, (b) skeleton load-displacement response, (c) secant stiffness, and (d) dissipated energy.

in Fig. 28(b) and (c), both the load-carrying capacity and the secant stiffness of the conventional SCPCs exceed those of the AP-SCPC. Additionally, Fig. 29(d) indicates that the SCPCs with non-perforated SPSWs demonstrate superior energy dissipation capacity compared to the AP-SCPCs. However, the perforations in the SPSWs notably enhance the self-centering capability of the SCPCs. For instance, the residual displacement of OP-1-0.4, caused by the out-of-plane buckling of the infill plates, is significantly reduced upon unloading compared to the conventional SCPCs. This observation highlights that SPSW perforations can mitigate out-of-plane buckling, providing additional benefits to the structural performance.

6. Conclusions

This paper introduced a novel self-centering concrete (SCPC) frame system incorporating post-tensioning tendons and steel plate shear walls with auxetic-shaped auxetic perforations (AP-SPSWs) as re-centering and energy dissipation devices, respectively. Repeated cyclic loading tests were conducted on three AP-SCPC subassembly specimens at a 0.5 scale to investigate the cyclic behavior of the newly-developed system. Finite element modeling was also performed for parametric studies on key design variables. The main conclusions drawn from this study are as follows:

- (1) The experimental results revealed that the failure modes of all specimens mainly concentrated on the “fuse” elements, i.e., AP-SPSWs, which involved out-of-plane buckling and local fractures at the four corners and/or at the ligaments between the AP-SPSW boundary elements and inner perforated infill panels in the infill panels.
- (2) The content of perforations (porosity) of the fuse elements was identified as a critical factor influencing the hysteresis behavior of the AP-SCPC frame. The specimen OP-0.85-0.4 exhibited the ultimate load capacity 76 % greater than that of the specimen OP-1-0.55, while the porosity of the former was 49.4 % higher than that of the latter.

- (3) All specimens demonstrated acceptable self-centering and energy dissipation capabilities. The beam, columns, and post-tensioned (PT) tendons remained largely elastic, while the AP-SPSWs exhibited plastic behavior during testing. A theoretical method was developed to accurately calculate the PT force, beam axial force, and column shear forces in the AP-SCPC system.
- (4) A detailed finite element (FE) model was developed for the AP-SCPC structure. Parametric studies focused on the effects of geometric configurations of the perforations in the AP-SPSWs and the initial PT force on the load-displacement response of AP-SCPC system. The results indicated that, an increase in the web thickness of the infill panel from 3 mm to 5 mm results in an enhanced bearing capacity and dissipated energy by 43 % and 61 %, respectively. Increasing the r_1/D parameter is likely an effective way to improve the hysteresis performance of AP-SCPCs. The parameter r_1/r_2 of 1 is recommended for the design of AP-SPSWs due to the presence of greatest bearing capacity and stiffness and moderate energy dissipation. Additionally, increasing the initial PT force contributed to a reduction of residual deformation and an enhanced self-centering capacity of the specimens.
- (5) This paper primarily examines the seismic performance of AP-SCPCs under low-cycle loading conditions. It is essential to integrate time-history dynamic simulations into future research endeavors to gain more in-depth insights into the seismic behavior of AP-SCPC structural systems. Furthermore, a more extensive array of parameterized studies should be conducted on the peanut-shaped perforated configurations to optimize their performance within perforated steel plate walls.

CRedit authorship contribution statement

Yazhi Zhu: Writing – original draft, Funding acquisition, Formal analysis, Conceptualization. **Jun-Hua Zhang:** Writing – original draft, Validation, Formal analysis. **Xiaoning Cai:** Writing – review & editing, Supervision, Investigation. **Peng Wang:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Xingwang Chen:** Writing – review & editing, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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